

Recommender Systems in Delivery Platforms: Challenges, Solutions and Learnings

Tutorial at 20th ACM Conference on Recommender Systems

Minneapolis, Minnesota, USA, September 28, 2026

Presenters: Marcel Kurovski · Paavo Camps · Raghav Saboo

Agenda

An overview of the tutorial schedule, divided into **foundations and core challenges** as well as **production-scale solutions**.

Part I: Foundations & Challenges

- 08:30 - **Delivery Platforms as Multi-Sided Marketplaces**
09:00
- Trade-offs between consumers, merchants, and couriers
 - Mapping offline ML metrics to online business KPIs

- 09:00 - **Challenges of Recommendations in Delivery**
09:40
- Locality constraints, real-time stock, & cold-starts
 - Multi-objective optimization & high-dimensional scale

- 09:40 - **Surfaces and Entities for Personalized Recommendation and Ranking**
10:00
- Discovery to checkout: homepage, stores, and cart
 - Diversity of entities: individual dishes to carousels

- 10:00 - **Coffee Break (30 min)**
10:30

Part II: Solutions & Future

- 10:30 - **Production Solutions for Delivery Recommendations**
11:45
- Item Representation with Semantic IDs
 - Retrieval with Two-Tower Embeddings
 - Multi-Task Learning for Ranking
 - Whole Page Optimization and Explore/Exploit Frameworks
 - Advanced Ranking Architectures
 - Generative Recommendations

- 11:45 - **Future Directions & Open Problems**
12:00
- Real-time personalization
 - Agentic recommenders
 - Interactive Q&A and audience discussion

About Us

We are seasoned practitioners working on **personalization across recommendation, ranking and search** across the different domains of our platform including restaurants, grocery stores, convenience, retail and more.



Marcel Kurovski

Staff Applied Scientist

Marcel Kurovski leads personalized recommendations across 30+ countries in restaurants, grocery, and retail. His work focuses on transformer-based store ranking, multi-objective optimization, and cross-domain transfer learning. He hosts *Recsperts*, a podcast on recommender systems, and has presented at PyCon DE & PyData, EuroPython, and the O'Reilly AI Conference. He holds an M.Sc. in Industrial Engineering from KIT.



Paavo Camps

Applied Scientist

Paavo Camps has spent four years on personalization and dish taxonomy systems at Wolt, spanning neural collaborative filtering, transformer-based ranking, and LLM-assisted product categorization deployed at scale. He holds an M.Sc. in Data Science from Aalto University and Université Paris Saclay.



Raghav Saboo

Staff Machine Learning Engineer

Raghav Saboo is the tech lead for search and personalization ML within New Verticals. He previously worked at Amazon on the first large language models and distilled models for Alexa AI, and as an ML consultant building zero-to-one solutions. He has published in WSDM, CIKM, SIGIR, and RecSys, and holds a Master's from Duke and a combined BEng/MEng from Imperial College London.

AI Disclaimer

We used AI assistants to create this presentation, for text and visuals.

We, the presenters, reviewed all output thoroughly and stand behind every slide.

If something seems unclear, please ask: we welcome discussion, questions, and comments.

Your questions come first

We welcome questions and give them priority over three hours of lecture on recommendations in the delivery industry.

Ask clarification questions directly, at any time.

Keep deeper questions and comments for the end of each session.



Get the slides
bit.ly/RecDel

Goals

Our tutorial should provide you with an introduction to the unique challenges for personalized recommendation and ranking in the delivery domain, showcase solutions and share learnings from our experience to invite for further exchange.

Unique Challenges

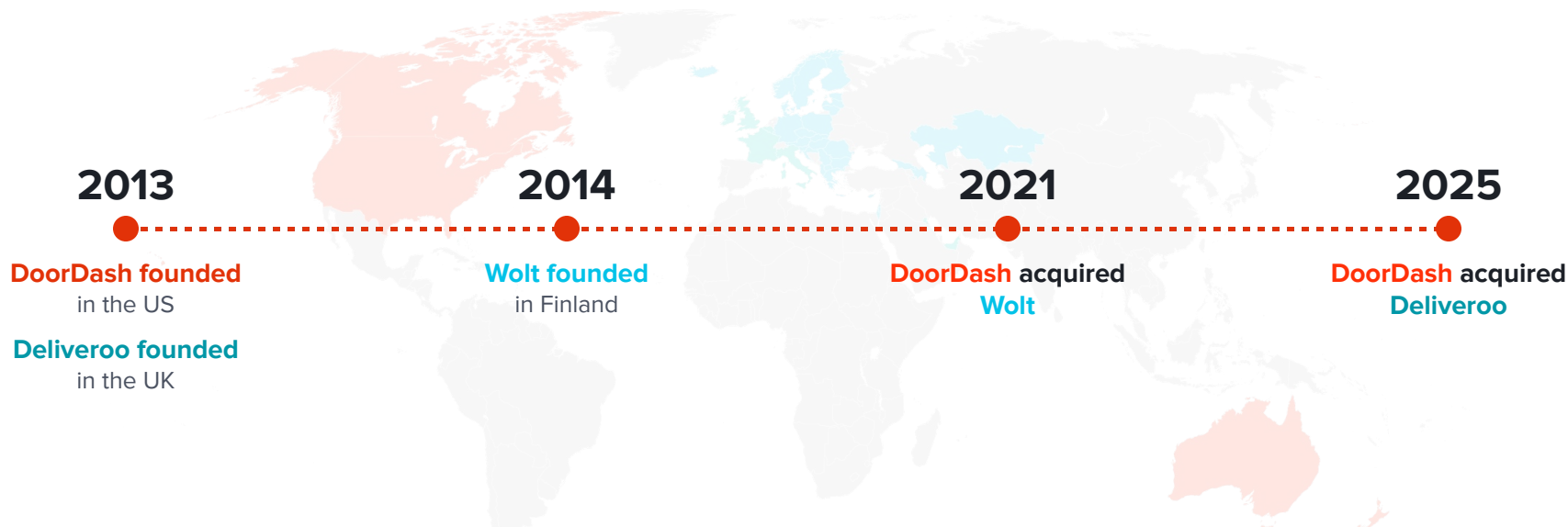
Show **what makes delivery recommendations hard**: locality and availability, short-term intent, three-sided cold start, competing objectives, and noisy merchant data.

Deployed Solutions

Share **how our production systems address these challenges** across the full recommendation stack, and what we learned.

Exchange & Outlook

Give newcomers an overview of the domain and practitioners a forum to **exchange experience and discuss future directions**.



Today, we serve more than **56M MAUs** in **40+ countries** around the world as one global company across three brands.

Session 1:

Delivery Platforms as Multi-Sided Marketplaces

Presenters: Paavo Camps

Stakeholder 1: The Consumer

The Goal Serving convenience and quality.

High Intent

“I want my favorite meal, and I want it fast.”

Discovery

“I want to browse a large catalog and find something new.”

What This Means for RecSys Sensible grouping, easy search, and serendipitous discovery.

Stakeholder 2: The Store (Merchants)

The Goal Drive marginal revenue and acquire new customers.

More Ways to Grow

Discovery brings in customers beyond the store's existing audience. Deals and ads offer additional ways to attract demand.

Control Over Demand

Commissions and delivery demands affect margins; a busy store may need to pause orders during peak hours.

What This Means for RecSys Give merchants a fair chance to be discovered while respecting availability and capacity

**Stores handling
their own delivery.**

**B2B
Delivery-as-a-Service.**

**Ghost Kitchens solely
on the platform.**

**SevenRooms -
reservations.**

Stakeholder 3: The Courier

The Goal Maximize earnings efficiently.

Ideal Tasks

Short distances, high pay, bundled orders, optimal drop-off locations.

Employment Models

Per-task gig work vs. full employment (varies by jurisdiction).

What This Means for RecSys Delivery time and courier cost can change which recommendations are best. Ranking should account for both through ETA and courier pricing signals.

Stakeholder 4: The Platform

The Goal High efficiency, sustainable revenue growth, and long-term profit.

In Theory

The platform doesn't optimize for any single order — it optimizes for the marketplace as a whole: efficiency, margins, and growth over time.

In Practice

Every lever meant to steer those outcomes — pricing, incentives, routing — acts on the world by changing what gets ranked and shown.

What This Means for RecSys RecSys is a lever to shape demand and balance the marketplace.

Market place trade offs

Growth
vs.
Profit

Fulfillment Quality
vs.
Cost

GOV
vs.
delivery quality/speed

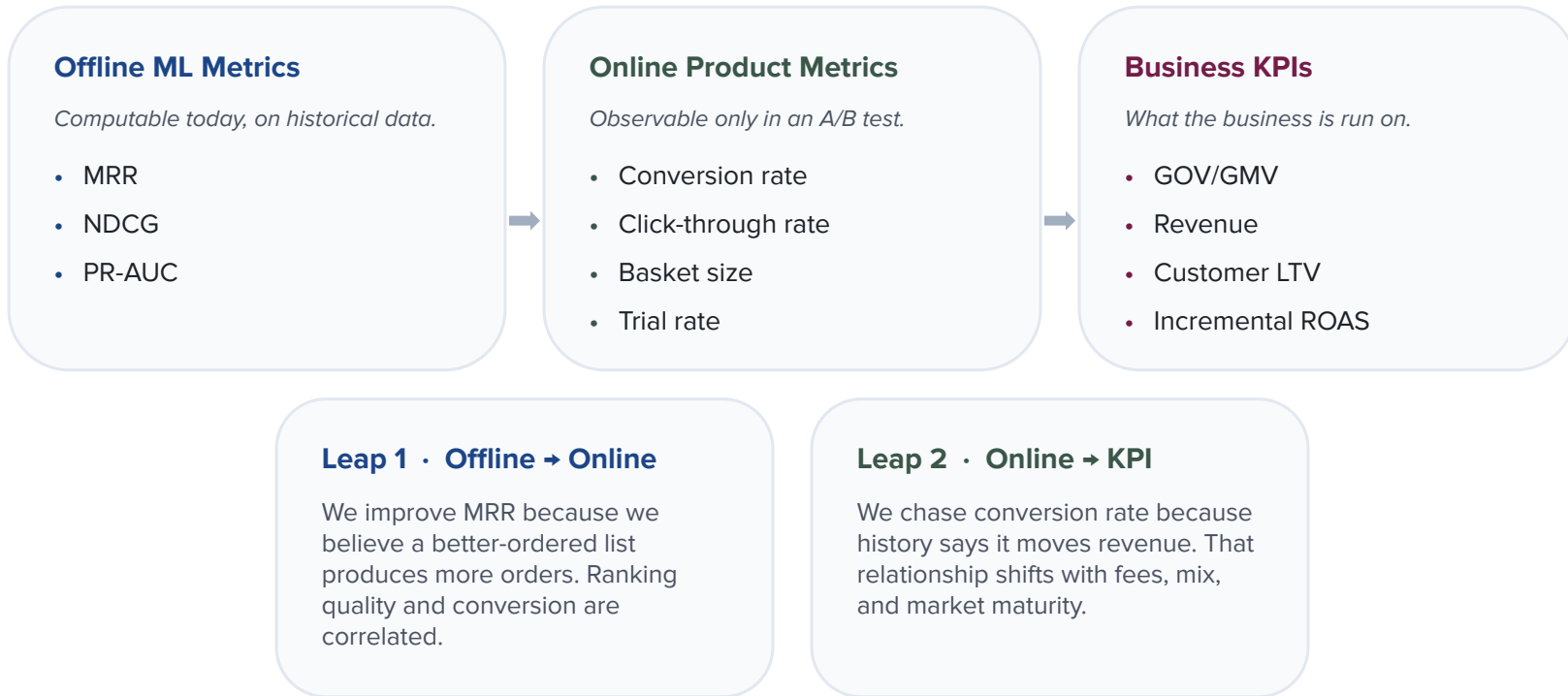
Highest Quality
Fulfillment Rate
vs.
its own components

THE METRIC TRANSLATION CHAIN



“Profit” is not fully differentiable.

Three Layers, Two Leaps



Chain 1 — Core Conversion



The Default Chain

- ✓ **Clean:** one offline target, one online target, one business outcome.
- ✓ **Measurable:** every link has an established, well-instrumented metric.
- ✓ **Finite:** and this is the problem — it eventually stops paying out.

Chain 2 — Exploration and Lifetime Value



Same Machinery, Different Target

- ✓ **Offline:** MRR computed only on stores or items this user has never bought.
- ✓ **Online:** store trial rate, first-time cross-category purchase rate.
- ✓ **Business:** retention and lifetime value rather than this session's revenue.

Ranking by Expected Value

score = **P(conversion)** × **expected commission or contribution**

Predict what the click is worth, not just whether it happens.

The shift

From probability of an order to the value of an order. Two candidates with the same click odds are no longer interchangeable.

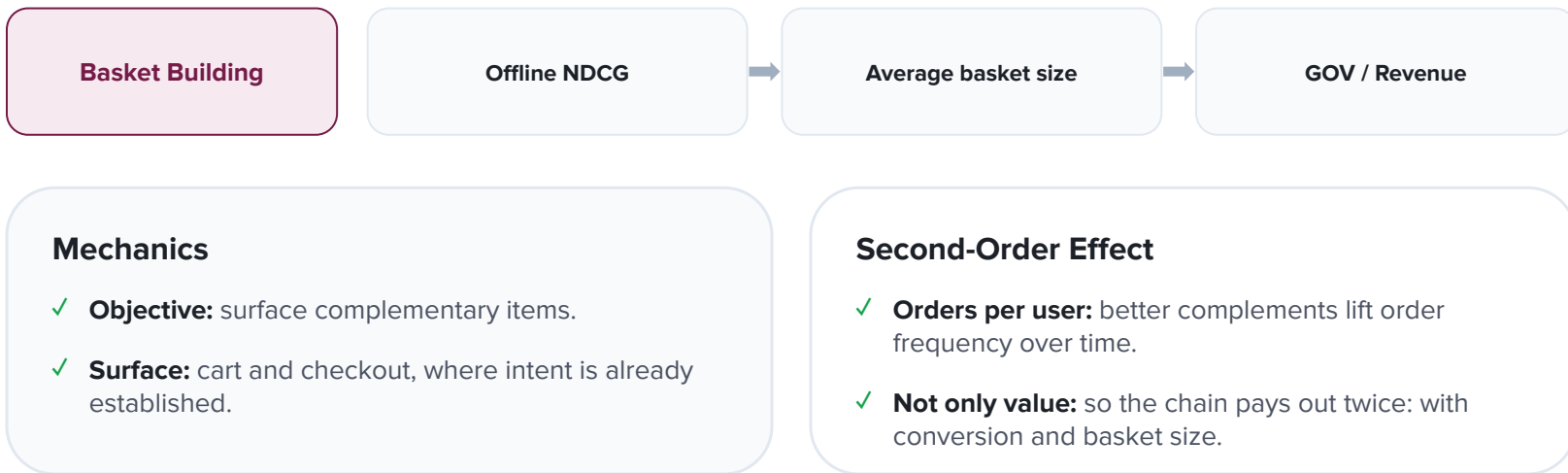
The effect

A slightly less likely but far more valuable order can outrank a safe one.

The risk

Value weighting is exactly where business thumbs land on the scale.

Basket Building Is a Separate Goal With Its Own Chain



The Same Chain for Ads



$$\text{ad_score} = \text{quality_score}^{\alpha} \times \text{effective_bid}$$

Generalized second price — the winner pays a score-ratio-adjusted second price, instead of their own bid.

Ad Score Mechanics

- ✓ **Quality score:** skews toward predicted conversion for CPA campaigns and predicted click-through for CPC.
- ✓ **The α exponent:** controls how much relevance can outweigh willingness to pay — the main dial between users and advertisers.
- ✓ **Clearing price:** a score-ratio-adjusted second price.

Last but not least:

Category	Metrics	Description
Coverage	Effective catalogue size	How much of the catalog gets recommended
Diversity	Intra-list diversity	How different the items in one recommendation list are
Serendipity	Unexpectedness, serendipity	Whether users discover relevant items they might not have found otherwise
Fairness	Exposure by creator or group, outcome gaps across user groups	Whether recommendations and their benefits are distributed equitably
User experience	Retention, satisfaction surveys, hide or report rate	Whether recommendations remain valuable over time
System performance	Latency, throughput, serving cost	Whether the system can deliver results reliably at scale

Session 2:

Challenges of Recommendations in Delivery

Optimizing under locality, availability, multi-sided marketplace, and semantic constraints.

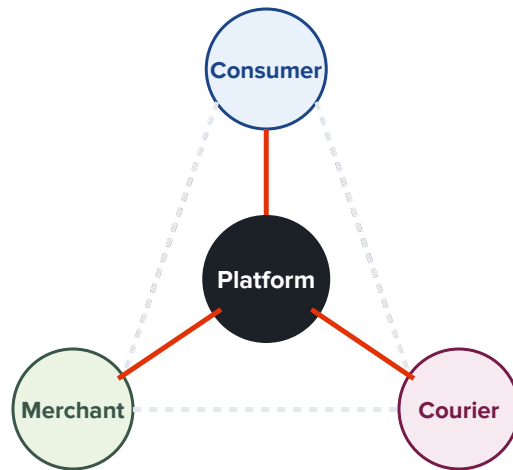
Presenters: Raghav Saboo

Delivery Recommendations Require Multi-Stakeholder Optimization Under Dynamic Local Constraints

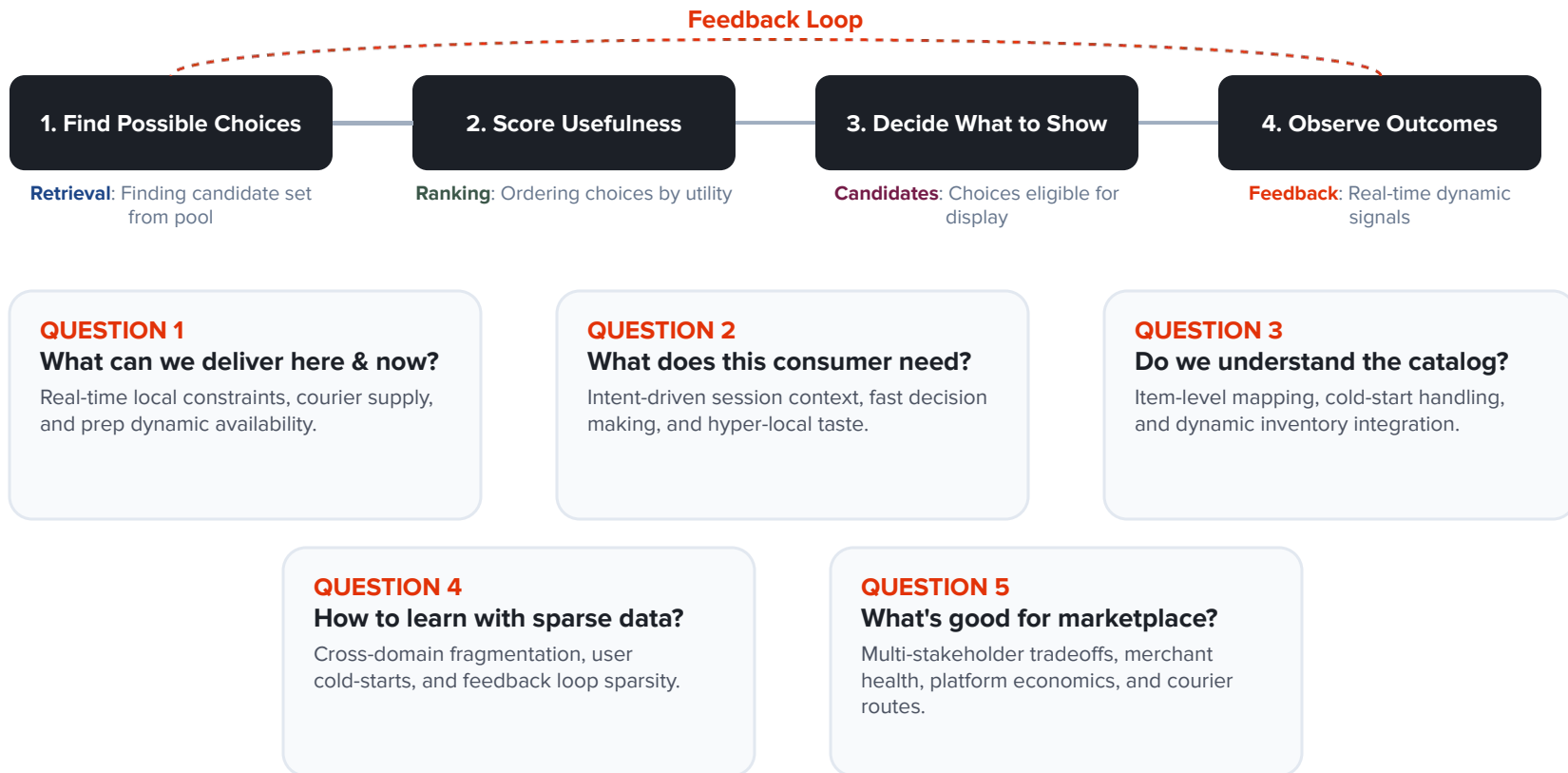
System Constraints & Real-Time Dynamics

- Unlike traditional e-commerce platforms with global warehouses, delivery recommendation systems operate under strict physical and temporal limitations.
- The marketplace is inherently multi-sided, requiring simultaneous optimization for consumers seeking relevance, merchants managing capacity, couriers needing efficient routes, and platform economics.
- Furthermore, user intent is highly mission-driven, and cross-domain fragmentation creates persistent cold-start challenges across the network.

Multi-Sided Marketplace Network



Five Questions Behind Every Delivery Recommendation



The Available Choices Change with Location and Time

@ MAIN TAKEAWAY

Preference alone does not determine which recommendations are useful.

Illustrative Request: "Dinner Delivered to 742 Evergreen Terr."

Evaluating candidate pool for a specific order context at 7:00 PM evening peak

RESTAURANT	REAL-TIME STATUS	POOL ELIGIBILITY
Burger Haven	Outside delivery zone polygon	Hard Exclusion
Pasta Bistro	Store closed at 6:30 PM	Hard Exclusion
Ramen House	Requested dish sold out	Hard Exclusion
Taco Supremo	High prep time (55 mins)	Rank Penalty
Curry Express	Open, in-stock, 20 min prep	Optimal Candidate

Store radius & delivery area polygons

Active operating hours

Item-level stock levels & prep times

Dynamic Candidate Set Shift

LOCATION A (7:00 PM)

1. Curry Express

2. Taco Supremo

3. Noodle Bar



LOCATION B / 7:15 PM

1. Burger Haven

2. Pizza Lounge

~~Curry Express~~

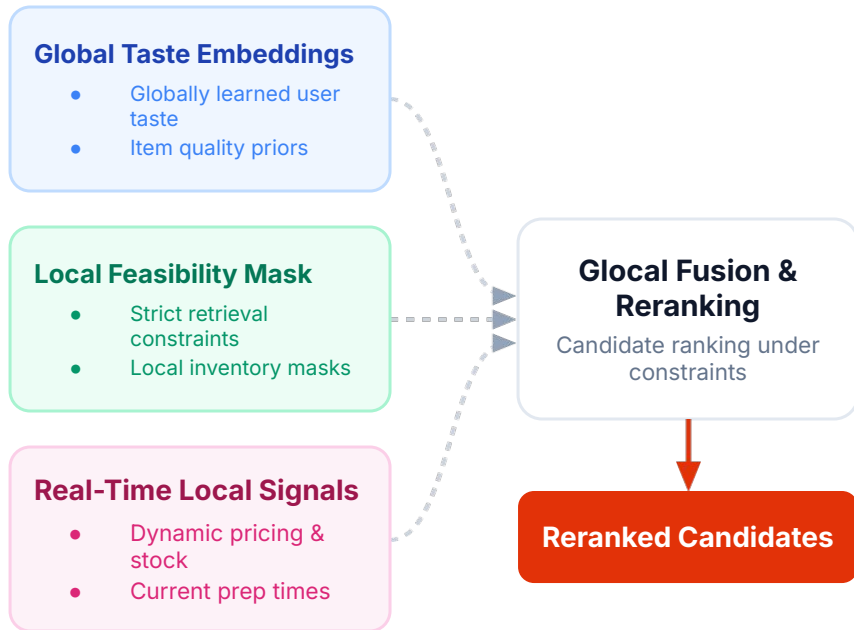
Uncertainty Challenge

Recommendation engines require real-time freshness checks at query moment and must fail gracefully.

Restricted strictly to deliverable items

Dynamic ranking feature vs static attribute

Glocal Architectures Fuse Global Taste Priors with Real-Time Local Feasibility



System Operations

Search & Discovery

Models across surfaces like **store pages**, **global search**, **cart**, and **checkout** seamlessly leverage global user taste and quality priors during initial phases.

Catalog Integration

Catalog linking maintains real-time local inventories to guarantee maximum accuracy and uphold strict service level agreements (SLAs).

Delivery Search is Mission-Driven and Requires Dual-Intent Modeling



User Profile State

Dynamic split intent

Persistent Memory (Long-Term)

- ✓ Favorite Stores & Past Orders
- ✓ Dietary Constraints & Preferences
- ✓ Household Size & Delivery Location

Session Intent (Short-Term Mission)

- ✓ Urgent Meal Planning (ASAP dinner)
- ✓ Stocking Pantry & Fridge Staples
- ✓ Casual Browsing / Impulsive Snacking

70%

Broad Categorical Queries

90%

Initiate Session Without Exact Product

Need for Disambiguation

Because consumer intent is fluid, a static single user embedding fails to capture instant state shifts.

Dual-intent modeling allows the system to decouple long-term baseline preferences from real-time session needs, driving precise query disambiguation.

Industry Search Analysis

Data highlights that standard search is heavily exploratory. Up to 70% of all queries focus on high-level categories rather than explicit brands, and 90% of sessions begin with open-ended browsing.

[Amazon Sponsored Brands guide, page 5](#); [Google/Ipsos, August 2015, reported on page 8](#).

Local Marketplaces Suffer from Interacting User, Store, and Item Cold Starts

New User Cold Start

No Order History

- Lacks historical transaction and preference signals
- Relies heavily on neighborhood and localized cohort priors
- Requires active exploration strategies to map user tastes

New Store Cold Start

No Engagement Metrics

- Lacks conversion, rating, and dispatch metrics
- Leverages geographic similarity and cuisine taxonomies
- High operational uncertainty (prep time, dispatching)

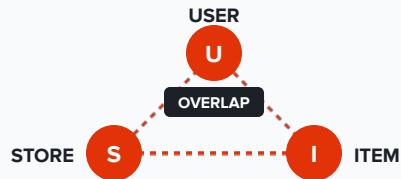
New Item Cold Start

Delayed Feedback

- Rapid catalog rotations with sparse immediate feedback
- Uses textual and visual catalog embeddings to map similarity
- Balance exploration with conversion-focused placements

Cross-Vertical Behavioral Silos

Existing users transitioning between distinct domains (grocery, retail, restaurant delivery) create localized cold starts due to isolated historical profiles. Bridging these behavioral silos is critical for sustaining overall marketplace growth and network liquidity.



Unstructured and Inconsistent Catalog Data Bottlenecks

Semantic Retrieval



**Raw Merchant
Data**



AI Enrichment



LLM Translation



Semantic IDs



Better Retrieval

Catalog Data Chaos

CHALLENGE

Unstructured merchant inputs

- Noisy and inconsistent titles
- Missing or messy attributes
- Lack of standard taxonomy

AI Normalization

PROCESS

Standardizing representations

- Extraction & spell-checking
- Multilingual LLM localization
- Semantic ID auto-generation

The Performance Cap

IMPACT

Data quality dictates accuracy

- Flawed data bounds model utility
- Clean IDs linking unlock exact matches
- Better candidate sets for ranking

Sustainable Recommendations Require Active Exploration and Objective Blending

Balancing Conversion & Sustainability

Practical recommendation systems must balance short-term conversion with long-term ecosystem sustainability.

- ✓ **Page Diversity Bounds:** Optimizes for immediate conversion while actively preserving merchant diversity to prevent monopoly effects.
- ✓ **Active Exploration Policies:** Implements time-decaying popularity boosts for new merchants, ensuring new entrants receive a fair chance to build audiences.
- ✓ **Seamless Sponsored and Promotion Blending:** Checkout recommendations drive incremental basket size while blending organic, sponsored, and exploration rankings to preserve consumer trust.

Example of Surface-by-Surface Objective Matrix

How recommendation targets adapt dynamically across high-intent touchpoints.

	Conversion Focus	Sustainability / Exploration
1. Home Page Discovery feed	Conversion & LTV Personalized relevance	New Entrant Boost Improve Cold Starts
2. Search Results Active intent matching	Organic Ranking Exact keyword match	Merchant Diversity Improve Discoverability
3. Cart Screen Pre-checkout consideration	Sponsored Placement High bid & context ads	Objective Blending Mix organic & sponsored
4. Checkout Final micro-purchase	Basket Expansion Incremental items	Strict Relevance Preserve consumer trust

Constraint-Aware, Multi-Objective Architectures Drive Marketplace Success

Local First

Feasibility Constraints

- ✓ Real-time physical feasibility filtering
- ✓ Strict localized delivery constraints
- ✓ Hyper-local ranking architectures

Memory over Sessions

Persistent Preferences

- ✓ Seamless long-term user profiles
- ✓ Immediate short-term intent signals
- ✓ Dynamic dual-intent adaptive mapping

Semantics over Text

AI-Enriched Catalog

- ✓ Dense catalog understanding systems
- ✓ Structured semantic IDs
- ✓ Normalized taxonomy over raw text

Higher Conversion

Frictionless user search matching

Consumer Trust

Highly accurate & reliable delivery

Healthy Economics

Optimized marketplace dispatch

Rapid Scaling

Seamless expansion into new verticals

Session 3:

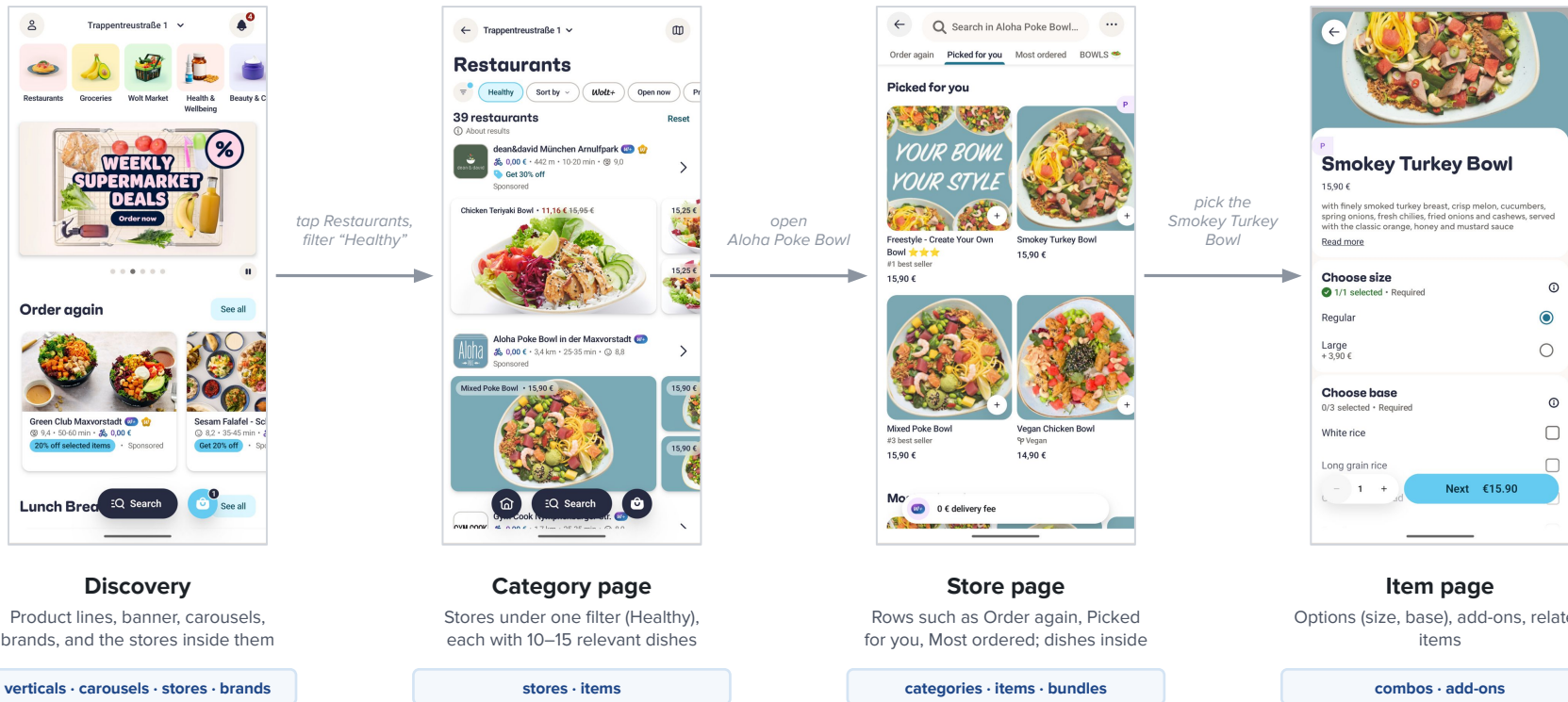
Surfaces and Entities for Personalized Recommendation and Ranking

Mapping the user journey from discovery to checkout across key recommendation surfaces.

Presenters: Marcel Kurovski

Our consumers browse different surfaces for a purchase

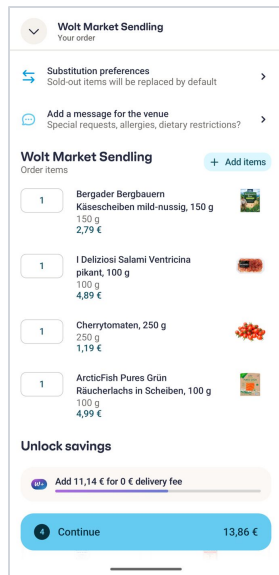
One restaurant order in Munich. Explorers walk more of these surfaces; decided consumers skip ahead.



Search enters the funnel sideways: a query returns stores or items directly.

Basket and checkout close the order and seed the next visit

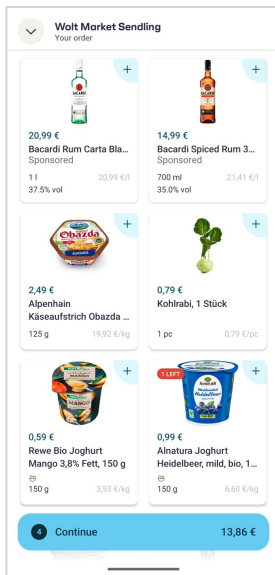
Shown with a Wolt Market grocery basket. A restaurant order has the same surfaces with smaller sets.



Basket

Selected items, substitution preference, savings nudge

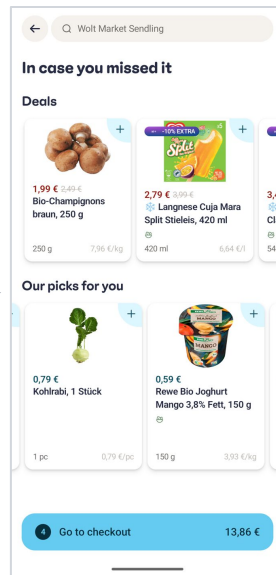
items · bundles



Basket cross-sell

Complementary products below the basket, incl. sponsored items

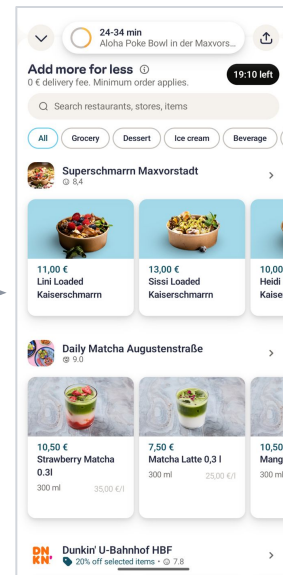
items



Checkout entry

"In case you missed it": deals and picks, one last small set

items



Post-order

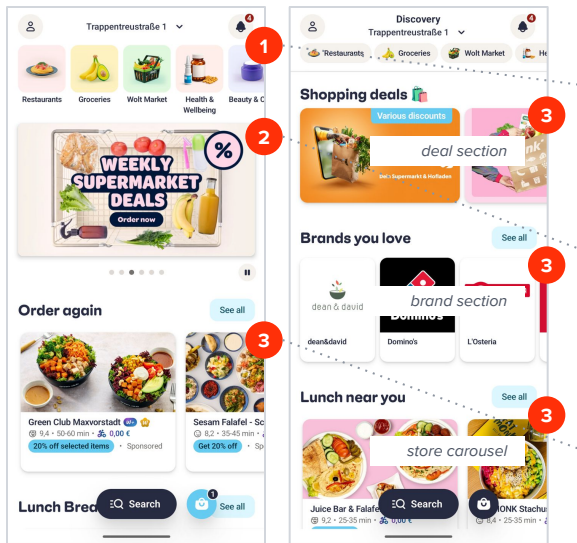
"Add more for less": store carousels with items for a follow-up purchase

stores · items

Search enters the funnel sideways: a query returns stores or items directly.

Discovery ranks stores into carousels, then the carousels

Three kinds of block on one screen. Each is first constructed (does it exist, where) and then ordered (what is inside).



1 Product line navigation

Tiles: Restaurants, Groceries, Wolt Market, Health & Wellbeing, Beauty, ...

construct · which tiles show

order · tiles in the slate

2 Hero banner

Campaign or editorial slate, e.g. "Weekly supermarket deals". One slate, several campaigns rotate.

construct · which campaign

order · rotation

3 Carousels

Horizontally ranked stores, from one or several verticals.

Store carousels: Order again · Recommended for you · Lunch near you · Burgers, Pizza

Brand sections: Brands you love (dean&david, Domino's, L'Osteria)

Deal sections: Shopping deals · Lunch Break Deals

construct · which rows, where

order · stores inside each

Challenge

Horizontal ranking: order the stores inside each carousel.

→ S4 advanced ranking architectures

Challenge

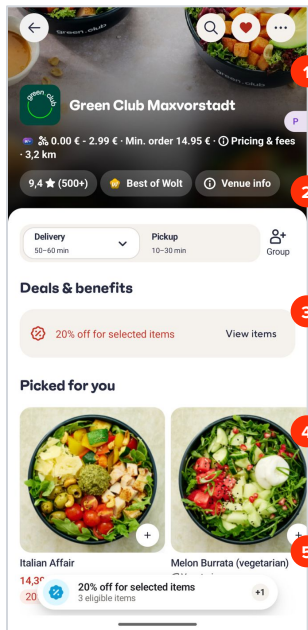
Vertical ranking: order the carousels on the page.

→ S4 whole-page optimization

First the candidate set, then the carousels. Location, delivery constraints, and opening hours define the delivering stores (glocal). Each store is scored and checked for eligibility per carousel, then the page is assembled: deduplication, ads injection, and the vertical ranking of the carousels. **A ranking of rankings.** ↪ S2
locality and availability

A store page is Discovery in miniature: header, rows, cards

One store's catalog inside the same layout family. A restaurant and a dark store fill it very differently.



Restaurant · Green Club Maxvorstadt

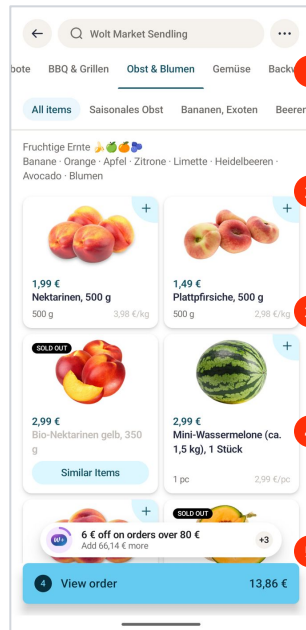
Store header Rating 9,4 (500+), fee, minimum order, distance, “Best of Wolt” badge. Store-level facts that change by the hour.

Delivery vs. pickup, group order 50–60 min delivery, 10–30 min pickup; a group-order entry.

Deals & benefits Promotions on selected items, a bundle-like unit shown before the menu.

Picked for you A ranked row of dishes inside one store. Further rows: Order again, Most ordered, then the merchant's menu categories (BOWLS, ...) in the merchant's order.

Dish cards Name, price, photo, badges such as “#1 best seller”. Merchant free text, no shared schema.



Grocery · Wolt Market Sendling (1P)

In-store search The primary navigation once the catalog exceeds a few hundred products.

Aisle navigation Merchant taxonomy in two levels: Obst & Blumen → Saisonales Obst, Bananen, Beeren ... Deep, merchant-specific.

Product cards Weight or pack size, price and unit price (€/kg); brand where one exists.

Sold out → “Similar items” Substitution surfaced in the UI: a recommender whose objective is acceptance.

Basket bar 4 items, 13,86 €: assembled over many loops. Further up: “Recently bought items” and “Deals” rows.

tens–hundreds of dishes ↔ tens–thousands of products · one meal ↔ a basket · kitchen prepares ↔ picker collects (substitution) → dishes vs. products

The app's screens hold many heterogeneous entities

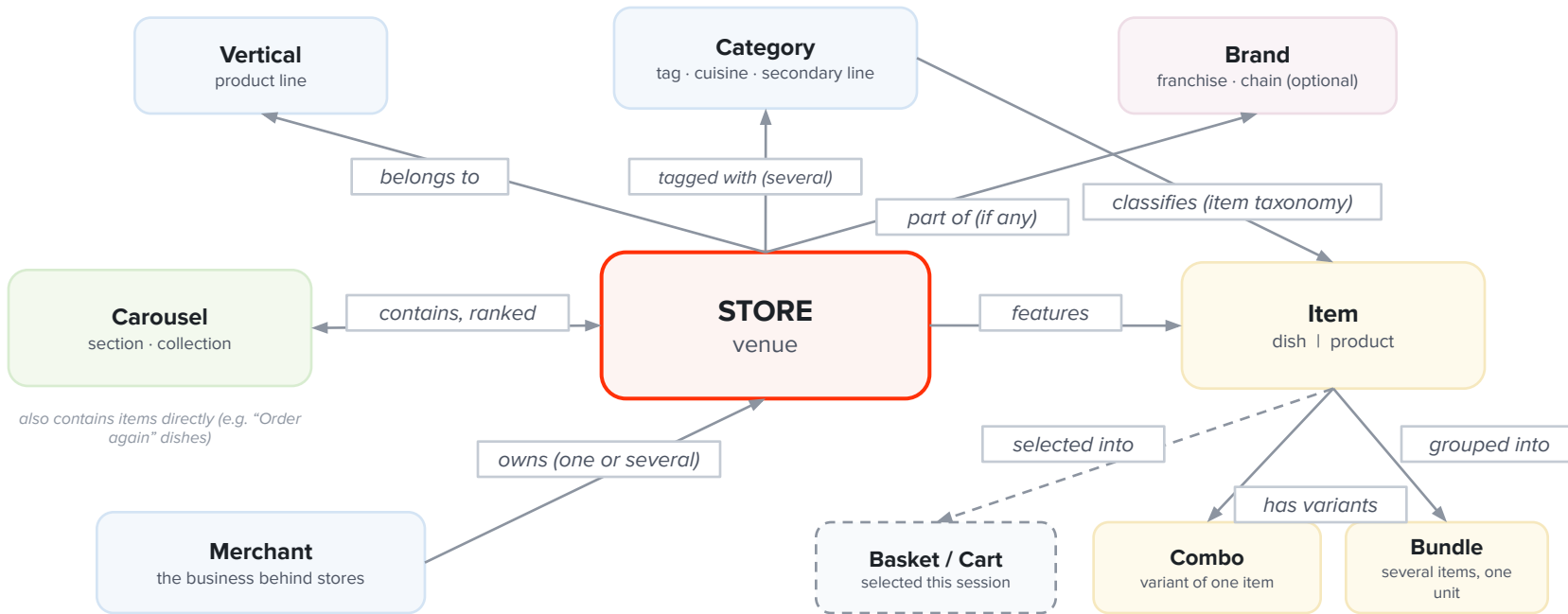
The main entities that occupy our app. They differ widely, so each needs its own personalized display.

Entity	Also called	Definition	Example
Vertical	product line	A domain of stores. Decided at the product line navigation.	Restaurants, Groceries, Wolt Market, Health & Wellbeing
Category	tag, cuisine, secondary line	A finer grouping inside a vertical. A store can carry several.	"Healthy", "Burgers", "Pizza"; "Supermarket", "Bakery"
Brand	franchise, chain	Several stores under one name. Preference forms here, if a brand exists.	McDonald's, Starbucks; Wolt Market
Store	venue	A place to order from, branded or independent: hours, radius, stock, ETA.	Wolt Market Sendling (1P); Aloha Poke Bowl (3P)
Item	dish · product	What a store sells: a dish (restaurant) or a product (any other store).	Smokey Turkey Bowl; cherry tomatoes 250 g
Combo	option, variant	One item in one configuration chosen by the consumer.	Smokey Turkey Bowl: regular or large; rice base
Bundle	deal, multipack	Several items arranged and sold together. Ranked as one unit.	"Buy 2 selected protein products, save 20%"; bowl + drink
Carousel	section, collection, slate	A themed, ranked list of stores or items (curated, rules, or generated).	"Order again", "Brands you love", "Lunch Break Deals"

Most of these entities are ranked somewhere in the app. Verticals in the product line navigation, categories and brands as tiles and rows, stores inside carousels and category pages, items inside stores, combos inside items, bundles next to items.

Everything connects to the store

The same entities as a relationship graph. Read it as a graph for modeling, as a tree for page construction.



Read top-down and the graph is a containment tree. A carousel contains ranked stores, a store contains ranked items, an item contains ranked combos. A page is a ranking of rankings. → P4 whole-page optimization

Item ranking differs between Restaurants and New Verticals

Dishes in restaurants, products/items in New Verticals such as grocery, retail, and pharmacy.

	Dishes (Restaurants)	Products / Items (New Verticals)
Catalog size per store	tens to hundreds	tens to thousands
Authoring and structure	merchant free text, no shared schema	partial GTIN / barcode coverage, brand, size
Sub-item structure	combos and add-ons (regular or large, rice base, extra side)	variants and pack sizes (1 L vs 1.5 L, 6-pack)
Repeat purchase	moderate	very high for staples
Substitutability	low within a restaurant, high across restaurants	high within a store: brand, size, and pack swaps
Consumer goal	choose one meal	assemble a basket of products/items
Fulfilment	kitchen prepares exactly what was ordered	picker collects; out-of-stock items need a substitute
Ranking objective	which dish converts the store visit	basket relevance, basket size, substitution acceptance

Bundles are ranked as units built from independently ranked items. Nesting again.

Substitution is a recommender whose objective is acceptance, because the sale already happened. Same catalog, different loss function, different failure cost.

Three kinds of data describe every entity

Content, behavior, context, for every node in the graph. An inventory only; fusing them is Session 4.

Content metadata

what the entity is

- **Text:** names, descriptions, menu and aisle labels, tags, categories
- **Image:** dish photos, product packshots, store covers
- **Structured:** price, opening hours, delivery radius, allergens, rating, store statistics

Behavioral

what consumers did with it

- **Impressions:** shown in a carousel, and at which position
- **Clicks / visits:** store and item page opens
- **Purchases:** orders, reorders, basket adds and removals
- **Favorites:** saved stores and items
- **Ratings:** store and delivery rating after the order
- **Returns:** refunds for quality or delivery issues, and for failed or poor substitutions

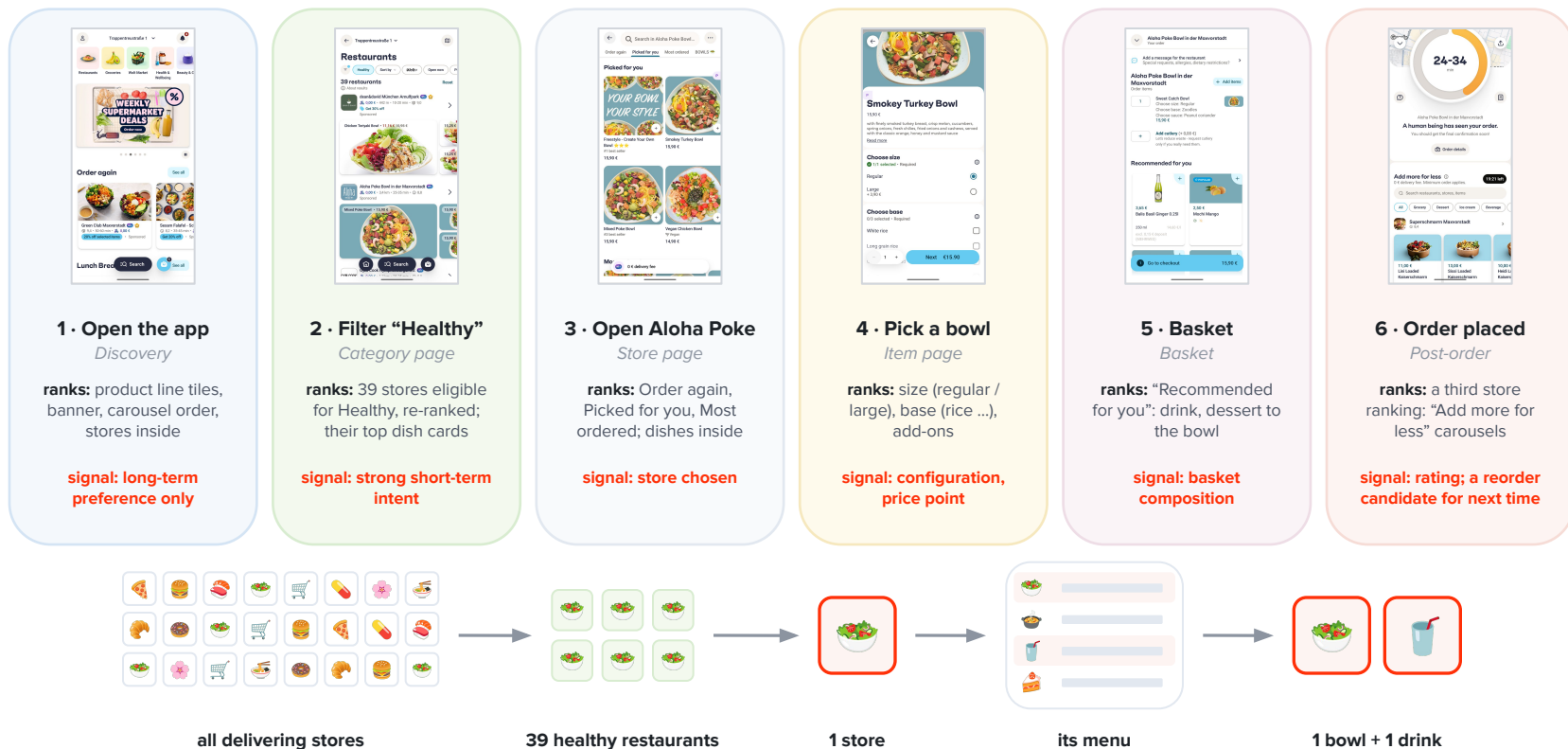
Contextual

the situation of the request

- **Time:** hour, weekday, holiday, mealtime
- **Location:** address, delivery distance, estimated delivery time (incl. meal preparation under current store load)
- **Company:** group orders, household vs. alone
- Device, weather

One session narrows stores first, then dishes, hop by hop

Lunchtime, Munich. Stores are ranked three times and dishes at most hops: more than a dozen model calls.



Same graph, different center of gravity per vertical

The entity graph is identical across verticals. Where the decisive ranking happens moves down it.

	Restaurants	Grocery (Wolt Market 1P, third-party 3P)	Retail (pharmacy, electronics, pet, beauty)
Typical entry	Discovery → cuisine row, filter, or carousel	Discovery → Groceries tile → store list with products	Discovery → vertical tile → in-store search
Decisive entity	store	store/item	item , sparse
Store choice	<i>the</i> decision	near-trivial: few stores deliver here	constrained by local coverage
Item ranking	secondary, converts the store visit	<i>the</i> decision: basket assembly	discovery-heavy, low frequency
Items per order	one to a few	many	one to a few
Reorder share	moderate	high	low
Hardest sub-problem	store relevance under locality	basket completion, substitution	cold start, sparse local catalog

One store ranker and one item ranker, shared naively across verticals, under-serve someone. That is why cross-domain transfer in Session 4 is worth its complexity. → P4 cross-domain store ranking, advanced ranking architectures

Four things to carry into the solutions

- 1 Consumers browse different surfaces for a purchase.** Discovery constructs the sets; later surfaces mostly order a set the consumer already narrowed. → **S4 whole-page optimization**
- 2 Stores are the centerpiece of a graph** that is also a containment tree. A page is a ranking of rankings. → **S4 store ranking across verticals**
- 3 Item ranking differs between Restaurants and New Verticals:** choose one dish versus assemble a basket of products/items, with combos, bundles, and substitution below. → **S4 item recommendation and ranking**
- 4 The same three data buckets describe every entity:** content metadata, behavior, context. Coverage differs by entity and market. → **S4 learning an item taxonomy, semantic IDs**

After the break, Session 4 turns these into production solutions. Learning an item taxonomy and item representations (semantic IDs)

- store ranking across restaurants, grocery, and retail
- item recommendation and ranking with multi-task models
- whole-page optimization and explore/exploit
- generative recommendations.

COFFEE BREAK

Be back in 30 minutes

Presenters: Rocket

Session 4:

Production Solutions for Delivery Recommendations

Solving challenges with scalable models and systems adopted to the needs of different markets, catalog and business objectives.

Presenters: Raghav Saboo, Paavo Camps, Marcel Kurovski

Item Representation

Presenters: Raghav Saboo

Combining Item Identity with Semantic Generalization

Learned Item-ID Features (Exact Identity)

Warm-Item Memorization: Preserves exact behavioral patterns & nuances for high-volume items.

New-Item Coverage: Limited; breaks down under sparse interactions or cold-start items.

Cross-Merchant Sharing: Siloed by merchant-specific schemas; no native ID transfer.

Cardinality: High catalog fragmentation and rapidly expanding vocabulary sizes.

Content-Derived Representations (Semantic Prior)

Warm-Item Memorization: Provides strong prior; may blur fine-grained item-level differences.

New-Item Coverage: Enables immediate coverage (quality depends on metadata richness).

Cross-Merchant Sharing: Maps equivalent items across disjoint merchant catalog schemas.

Key Limitations: Bounded by attribute quality, description depth, and loss objectives.

Coexistence Model: Linking Disparate IDs via Shared Semantic Embeddings

Merchant A · ID #1042
"Coca-Cola 12oz Can"
High interaction volume

Exact Item

Merchant B · ID #8891
"Coke 355ml Can"
Newly added listing (cold)

Semantic Related

Merchant A · ID #1043
"Diet Coke 12oz Can"
Distinct physical product

Item Meaning Comes from Content and Interaction Signals

1. Content Signals & Multimodal Encoding

Text & Attributes

- Brand & Size
- Catalog Taxonomy
- Product Descriptions

Product Imagery

- Package Appearance
- Prepared Dish Imagery
- Visible Label Info

Multimodal Encoders → Fused Dense Vector

Maps textual metadata and visual details into a joint embedding space to establish intrinsic item semantic priors.

2. Behavioral Signals & Supervision Objectives

Interaction Signals (Adaptation Path)

Search impressions, clicks, add-to-carts, completed orders, substitution sequences.

Supervision Objectives & Learning Mechanism

- **Clicks & Orders:** Teaches search relevance and purchase preference.
- **Substitutions:** Teaches functional item substitutability.

Note: Click and order supervision does not automatically capture substitution quality; requires dedicated replacement sequence objectives.

3. System Architecture: Global Item Representation vs. Live Operational State

Global Item Representation (Static Vector Space)

- Fused content embeddings capture catalog semantic prior.
- Behavioral adaptation aligns spaces for substitutability & preference.
- Portable across stores and catalog schemas.

Live Serving Layer (Dynamic Operational State)

- Store inventory levels & real-time stock status.
- Store opening hours, pricing, & promotions.
- Delivery feasibility & courier SLA filters (requires fresh serving state).

Residual Quantization Compresses Dense Vectors into Hierarchical Semantic IDs

■ The Residual Quantization Algorithm

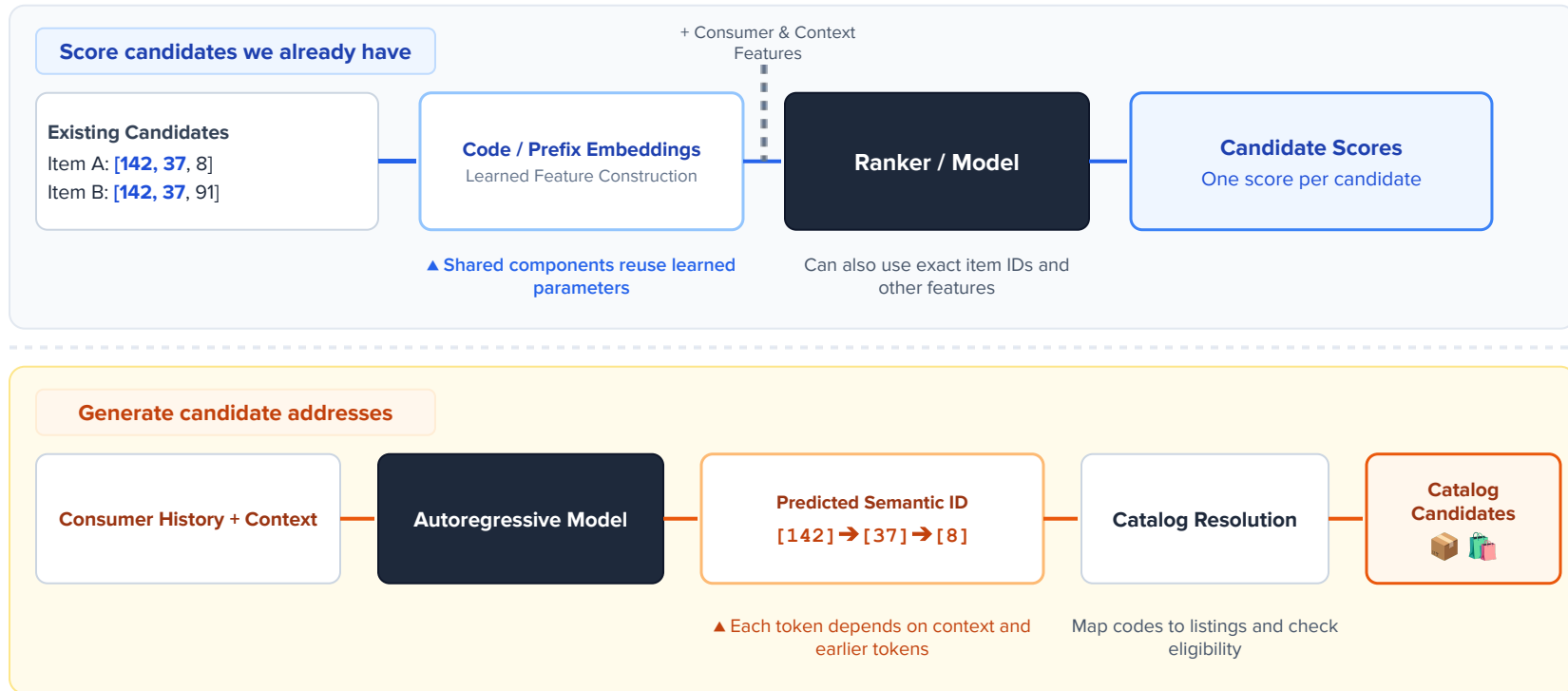
- 01 Continuous Vector Input**
High-dimensional continuous dense embeddings represent items.
- 02 Select Nearest Codeword**
Maps the input embedding to the closest codeword in active layer.
- 03 Isolate the Residual**
Subtracts the codeword from vector to isolate remaining signal.
- 04 Recursive Quantization**
Repeats process hierarchically (e.g., RQ-KMeans across 3 layers).
- 05 Generate Hierarchical IDs**
Outputs compact discrete code sequences to formulate semantic IDs.

■ Core Technical Properties

-  **Smaller Vocabulary Size**
Converts continuous embeddings into short, discrete code sequences (e.g., RQ-KMeans with 3 layers of 512 codes), drastically reducing vocabulary overhead.
-  **Robust Parameter Sharing**
The learned semantic hierarchy enables efficient parameter sharing between items, ensuring strong generalizations across overlapping item categories.
-  **Graceful Long-Tail Handling**
Niche products are grouped with their attribute neighbors in the lower layers of the tree rather than being lost to simple brand popularity skew.

The same semantic ID can be a model input or output

Ranking scores existing candidates. Generative retrieval predicts candidate ids.



Semantic IDs Mitigate Cold Starts but Require Strict Quantization Tuning

■ The Cold Start & Transfer Capability

Semantic representations naturally address catalog cold starts. By mapping immediately available content, new items are placed near known items in the embedding space.

- ✓ **Immediate Content Utility:** Uses rich textual and visual catalog metadata to position new items instantly without order data.
- ✓ **Rapid Merchant Onboarding:** Drastically reduces wait times, bootstrapping catalog discovery from day one.
- ✓ **Cross-Vertical Knowledge:** Successfully bridges behavioral silos by transferring embeddings across retail, grocery, and dining.

▲ Production Bottlenecks & Tuning

Deploying semantic IDs at scale introduces distinct production bottlenecks. Active monitoring and careful quantization tuning are crucial to sustain performance.

- **Codebook Collapse & Collisions:** Occurs when too many items map to a small set of codes, underutilizing representation bandwidth.
- **Semantic-Behavioral Mismatch:** When semantic content vectors conflict with actual transactional user behaviors and preference signals.
- **Retraining Instability & Tuning:** Requires prefix-adjusted quantization and taxonomy alignment to stabilize quantizer retraining.

Semantic IDs Trade Transfer Benefits for Quantization Complexity

■ Where They Can Help

- ✓ **New items:** Content provides features before interaction history exists.
- ✓ **New store listings:** Shared representations can reuse learning across catalogs.
- ✓ **Cross-domain transfer:** Possible when representations and tasks align, but requires empirical validation.

▲ What to lookout for

- **Code collapse:** Only a small part of the code vocabulary is actively used.
- **Full-sequence collisions:** Distinct items receive identical complete IDs.
- **Semantic-behavioral mismatch:** Content similarity may fail to align with task signals.
- **Code churn:** Retraining alters ID assignments consumed by downstream models.

Operational Guidelines: Monitor utilization & collisions • Evaluate cold-start & long-tail slices • Version quantizers • Coordinate dependent models & indices

Multi-Modal Signals Build a Universal Understanding of Item Meaning

1. Text & Attributes

Standardized Metadata

- Brand, Name, & Size
- Catalog Taxonomy
- Product Descriptions
- Structured Encodings

2. Visual Metadata

Product Photography

- Packaging Packaging
- Prepared Dish Imagery
- Label Visual Details
- Visual Pattern Maps

3. Behavioral Signals

User Interactions

- Search & Impressions
- Clicks & Add-to-Carts
- Orders & Replacements
- Substitution Sequences

Content Encoders & Dense Vectors

By embedding structured attributes, descriptions, and visual metadata, content encoders map raw, multi-modal inputs into robust dense vectors. This framework successfully isolates linguistic semantic similarity across disparate catalog schemas.

Behavioral Training & Operational Isolation

Behavioral interaction training dynamically adapts vector spaces to capture actual substitutability rather than pure textual correlation. This cleanly separates global item knowledge from store-specific operational constraints and live inventory.

Residual Quantization Encodes an Item in a Short Code Sequence

Worked RQ-KMeans Quantization Diagram & Formulation

Recursive Quantization Equations:

$$r_0 = z \quad | \quad k_l = \operatorname{argmin}_k ||r_{l-1} - c_l, k||^2 \quad | \quad r_l = r_{l-1} - c_l, k_l$$

z : input dense vector • r_l : residual vector at level l

c_l, k_l : codeword k in level- l codebook • k_l : selected codeword index

Level 1 ($l = 1$): Start with $r_0 = z$. Match nearest codeword c_1, k_1 in C_1 .

→ Selected Code: k_1 | Compute Residual: $r_1 = r_0 - c_1, k_1$

Level 2 ($l = 2$): Quantize residual r_1 . Match nearest codeword c_2, k_2 in C_2 .

→ Selected Code: k_2 | Compute Residual: $r_2 = r_1 - c_2, k_2$

Level 3 ($l = 3$): Quantize residual r_2 . Match nearest codeword c_3, k_3 in C_3 .

→ Selected Code: k_3 | Final Residual: $r_3 = r_2 - c_3, k_3$

Resulting Hierarchical Semantic ID: $[k_1, k_2, k_3]$

Key Technical Implications

1. Compact Vocabulary

With 3 level-specific codebooks of 512 entries, only 1,536 code vectors are maintained, expressing up to 512^3 (~134.2M) possible discrete code sequences.

(Numerical configuration is illustrative)

2. Parameter Sharing

Semantically related items share prefix components (e.g., k_1 or $[k_1, k_2]$), enabling structured parameter re-use across catalog items.

3. Compression Tradeoffs

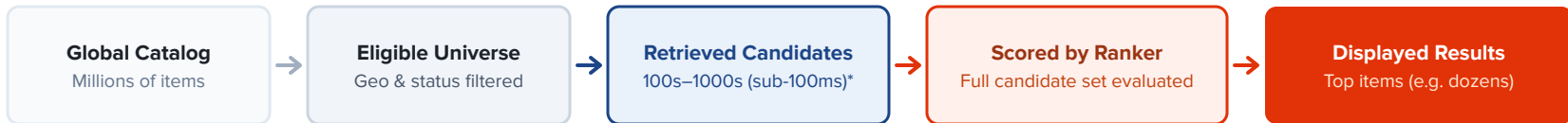
Quantization involves tradeoffs: different items may collide to identical code sequences or lose fine-grained distinctions based on codebook depth.

Note: Does not imply uniform code usage, collision-free IDs, guaranteed long-tail improvement, or human-readable taxonomy.

Two Tower based dense retrieval

Presenters: Raghav Saboo

Retrieval and Ranking Solve Different Parts of the Decision



* Candidate counts and sub-10ms targets are illustrative and vary by domain architecture.

■ Retrieval Focus

STAGE CHARACTERISTICS

- **Objective:** Candidate recall across the vast eligible universe.
- **Model Interaction:** Efficient matching via decoupled representations (e.g. Two-Tower ANN search).
- **Compute Profile:** Broad coverage using lightweight operations under tight latency guardrails.
- **Evaluation:** Eligible Recall@K and strict sub-10ms execution SLAs.

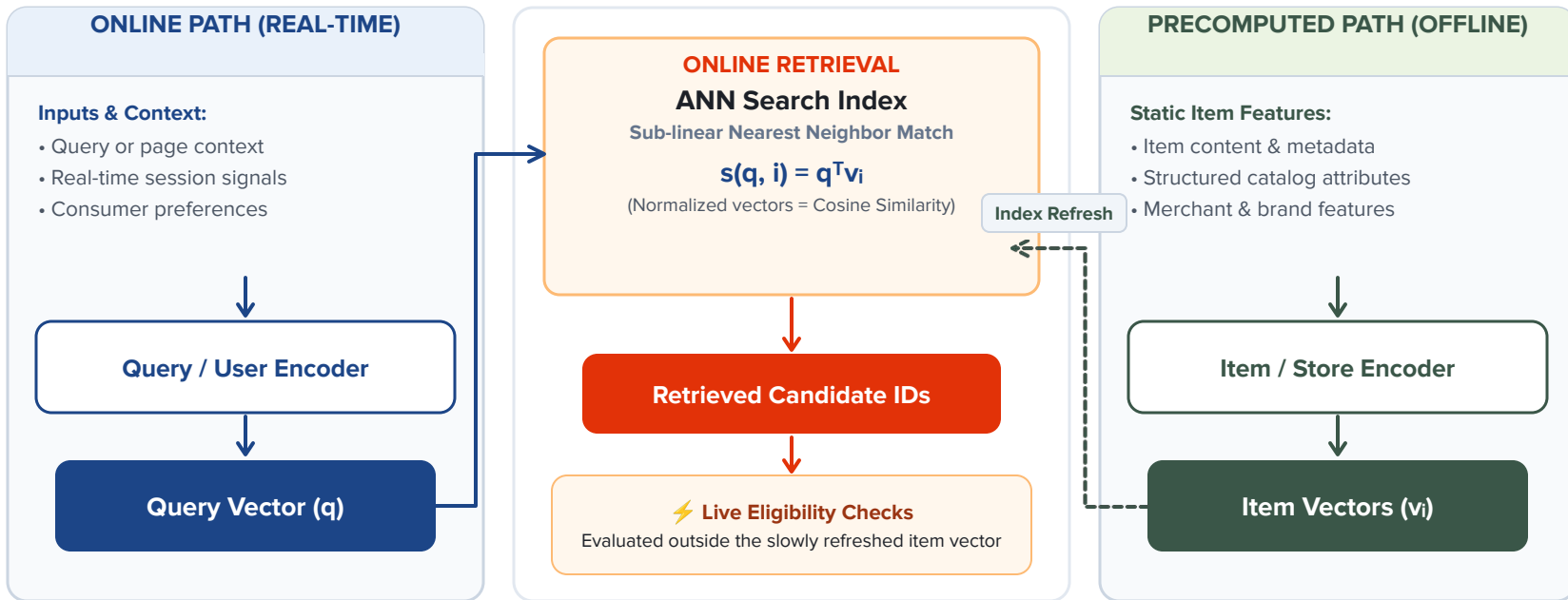
■ Ranking Focus

STAGE CHARACTERISTICS

- **Objective:** Quality of ordering, precision, and multi-objective blending.
- **Model Interaction:** Richer query–item joint features, real-time context, and cross-store signals.
- **Compute Profile:** More computation allocated per candidate over the retrieved scoring set.
- **Evaluation:** Ranking quality metrics (NDCG, mAP) and downstream conversion metrics.

Domain Qualification: Some store surfaces (e.g., single-store menus) have a small enough eligible set to score directly without retrieval, whereas large multi-store catalog spaces strictly require a retrieval stage.

Independent Encoders Make Retrieval Precomputable



Key Architectural Trade-Off: Independent encoding enables precomputation and fast search, but limits rich query–item interactions until later ranking stages.

Retrieval Learns from the Events and Comparisons We Choose

■ Behavioral Supervision & Event Signals

Impression (Baseline Visibility)

Exposure only; baseline visibility, not positive preference.

Click (Engagement Signal)

Frequent engagement; strongly influenced by position bias.

Add-to-Cart (High Intent)

Strong intent signal; does not guarantee purchase.

Order (True Conversion)

High-fidelity conversion; sparse and delayed downstream signal.



Key Principle

Training objectives determine which events serve as positive labels. Do not prescribe a universal increasing weight across all events.

■ Negative Sampling & Contrastive Loss

Sampled Contrastive Loss

$$\mathcal{L} = -\log \left[\frac{\exp(s^+/\tau)}{\sum_j \exp(s_j/\tau)} \right]$$

- **In-Batch Negatives:** Drawn from other examples' positive items in the same batch.
- **Hard Negatives:** Exposed-but-skipped items; informative, but remain noisy and position-biased.
- **False-Negative Risks:** Duplicate products or multiple plausible alternatives mislabeled as negatives.

Feasibility Constraints Belong Throughout the Serving Path

RETRIEVAL & SERVING-TIME FILTERING ARCHITECTURE

⚡ Dynamic Live Inputs

- **Delivery Reach:** Geo-polygons & radius
- **Store State:** Hours & pause status
- **Item Stock:** Store inventory count
- **Service/Courier:** Fleet availability

1. Pre-Filtering

(Before ANN Search)

- Query partitioning by location/zone
- Hard store eligibility & operating windows

2. Inline ANN

(During Vector Search)

- Bitset / metadata attribute filtering
- Evaluated directly within HNSW/index traversal

3. Post-Filtering

(After Retrieval)

- Exact real-time stock & price validation
- **Overfetch / Fallback:** Retrieve $K > N$ to prevent candidate fill drops

Prefiltering Tradeoff

Narrows search universe early for high efficiency; depends strictly on clean geographic/store partitionability.

Inline Filtering Tradeoff

Incorporates dynamic metadata directly into vector traversal; depends on specialized index support and index freshness.

Postfiltering & Overfetch

Supports exact current validation; requires overfetching candidates or trigger fallbacks to avoid leaving too few candidates.

💡 **Architectural Principle:** Slowly refreshed item vectors cannot by themselves enforce rapidly changing feasibility. Hard eligibility rules enter via store/service pre/post-filtering, while uncertain stock estimates or courier supply guide downstream ranking scores.

Multi Task and Multi Objective Ranking

Presenters: Raghav Saboo

Multi-Task Learning Optimizes Across the Entire Customer Engagement Funnel

Multi-Task Multi-Label Funnel Objectives



✗ Why Single-Task Formulations Fail

Optimizing individual steps in isolation leads to systemic inefficiencies:

- **CTR Over-Optimization:** Focusing purely on clicks generates clickbait. The model prioritizes eye-catching items that users frequently click but ultimately abandon without purchasing.
- **Extreme Data Sparsity:** Training exclusively on conversion (pConv) data is severely limited. Because orders are rare events relative to impressions, the model suffers from high variance and poor representation learning.

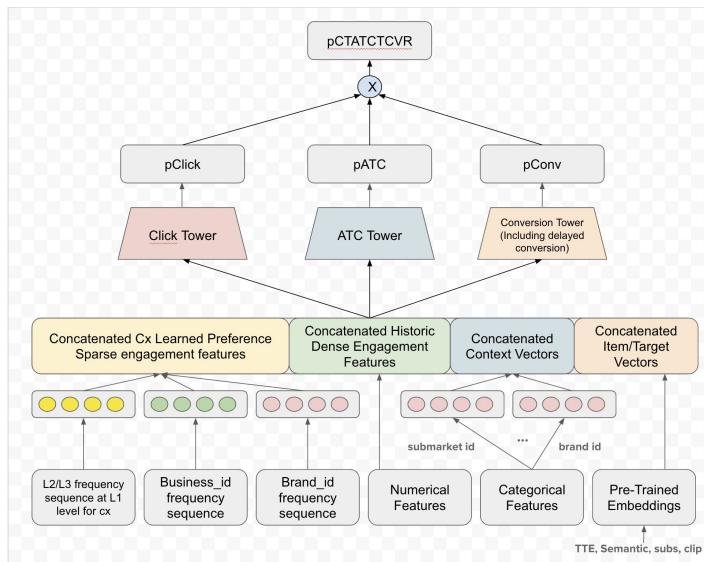
✓ The Multi-Task Learning (MTL) Advantage

Simultaneously training across multiple funnel objectives improves generalization:

- **Shared Representations:** Bottom layers share features across all tasks. This allows sparse signals (conversions) to benefit from dense signals (clicks and add-to-carts) during learning.
- **Balanced Output Predictions:** Model uses task-specific prediction heads to output probability vectors, balancing semantic relevance, user engagement, price affinity, and fulfillment logistics.

Advanced Multi-Task Routing Mitigates Negative Transfer Between Competing Objectives

▲ Multi-Task Learning Architecture



■ Solutions to the Seesaw Effect

The Seesaw Effect & Negative Transfer

Improving one task (e.g., conversion) degrades another (e.g., clicks) due to competing gradient updates. Advanced architectures reconcile these conflicting signals.

• Shared-Bottom Baselines

Branches a single shared trunk directly into separate task heads. This layout is highly vulnerable to destructive gradient interference.

• MMoE & PLE Routing

Utilizes separate, parameterized gating networks to select distinct mixtures of shared and task-specific experts dynamically.

• Calibrated Blending

Enables highly calibrated, task-specific blending of relevance, engagement, and conversion probabilities without performance degradation.

Production MTML Requires Careful Task Weighting and Surface Calibration

Loss Tuning

Task Loss Weighting

Balancing Competing Losses

- ✓ **Manual Tuning:** Statically assigned weights prioritizing immediate conversion over clicks.
- ✓ **Uncertainty Weights:** Adjusts dynamically based on task loss variances during training.
- ✓ **GradNorm:** Dynamic gradient normalization to balance different task training rates.

Calibration

Prediction Calibration

Strict Head Calibration

- ✓ **Isotonic Regression:** Non-parametric mapping to realign predictions to empirical rates.
- ✓ **Temperature Scaling:** Post-processing alignment of predictions across unique markets.
- ✓ **Market Blending:** Ensures predictions are fully calibrated before cross-prediction blending.

Surface Adaptation

Surface Specialization

Context-Aware Outputs

- ✓ **Search Results:** High semantic relevance matching targeted, high-intent user queries.
- ✓ **Homepage Carousel:** Prioritizes exploration, discovery, and merchant diversity.
- ✓ **Cart & Checkout:** Impulse-driven complementary additions to boost overall basket size.

End-to-End Evaluation Connects Offline Diagnostics to Marketplace Impact

Offline Diagnostics

Model Quality & Calibration

- ✓ **Head-Level AUC & Log-Loss:** Measures task-specific prediction accuracy
- ✓ **Expected Calibration Error (ECE):** Ensures predicted probabilities map to actual rates
- ✓ **NDCG & MRR:** Evaluates relative ranking quality of recommendations

Online Marketplace Metrics

A/B Test Guardrails & Business Impact

- ✓ **Conversion & Order Rate:** Drives overall marketplace transaction volume
- ✓ **Gross Order Value (GOV):** Tracks absolute dollar flow across the platform
- ✓ **Basket Subtotals:** Monitors user spending behavior and order sizes

STAGE 01

Representations

Semantic IDs

Dense item vectors capture taxonomy, intent, and relationships.

STAGE 02

Candidate Gen

Two-Tower Retrieval

Scalable dual-encoder matches query and candidate spaces.

STAGE 03

Precise Sorting

PLE / MMoE Models

Multi-task learning networks predict individual objective likelihoods.

STAGE 04

Marketplace Guard

Physical & Coherence

Real-time filtering enforces physical feasibility & whole-page layout logic.

Whole Page Optimization and Explore/Exploit Frameworks

Presenters: Paavo Camps

Building a homepage

Problems with Pointwise Ranking

Score each store on its own and sort by score

- ✓ **Heterogeneous units:** stores, carousels, grocery collections and ads compete for one page.
- ✓ **Unequal value:** a ~\$X grocery basket and a ~\$Y restaurant order are not the same click.
- ✓ **Feedback loop:** content near the top earns clicks, looks better in the data, keeps its slot.

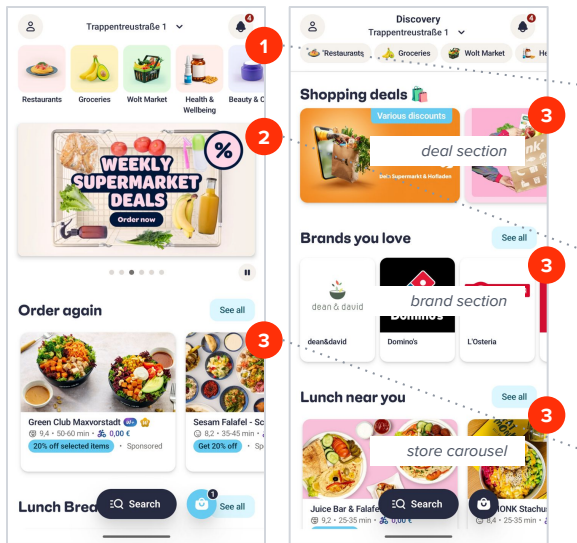
Whole-Page Optimization Improvements

The unit of optimization becomes the page instead of the item.

- ✓ **Construction under constraints:** diversity, deduplication, ad quality.
- ✓ **Reserved exposure:** explore/exploit keeps some slots for types of content that are new or uncertain.
- ✓ **Cohesion:** affinity to catalog attributes and collections shapes what the page is about.

Discovery ranks stores into carousels, then the carousels

Three kinds of block on one screen. Each is first constructed (does it exist, where) and then ordered (what is inside).



1 Product line navigation

Tiles: Restaurants, Groceries, Wolt Market, Health & Wellbeing, Beauty, ...

construct · which tiles show

order · tiles in the slate

2 Hero banner

Campaign or editorial slate, e.g. "Weekly supermarket deals". One slate, several campaigns rotate.

construct · which campaign

order · rotation

3 Carousels

Horizontally ranked stores, from one or several verticals.

Store carousels: Order again · Recommended for you · Lunch near you · Burgers, Pizza

Brand sections: Brands you love (dean&david, Domino's, L'Osteria)

Deal sections: Shopping deals · Lunch Break Deals

construct · which rows, where

order · stores inside each

Challenge

Horizontal ranking: order the stores inside each carousel.

→ S4 advanced ranking architectures

Challenge

Vertical ranking: order the carousels on the page.

→ S4 whole-page optimization

First the candidate set, then the carousels. Location, delivery constraints, and opening hours define the delivering stores (glocal). Each store is scored and checked for eligibility per carousel, then the page is assembled: deduplication, ads injection, and the vertical ranking of the carousels. **A ranking of rankings.** ↪ S2 locality and availability

Homepage Recommendations Require Multi-Level Entity Ranking and Coherence

■ Homepage Carousel Structure

Carousel Header

e.g., "Flu Season" collection banner and navigation

Collection Container

e.g., "Health Essentials" bento block or card area

Individual Store Items

e.g., Individual Vicks and NyQuil product cards

Flu season



Health Essentials

Stay



\$7.89

Vick's VapoRub
Original Cough S...
30 min • DashMart

\$3.59

Lil' Drug Store
Vicks Nyquil Seve...
30 min • DashMart

\$4.99

15% off
Purell
42 min

■ Whole-Page Coherence

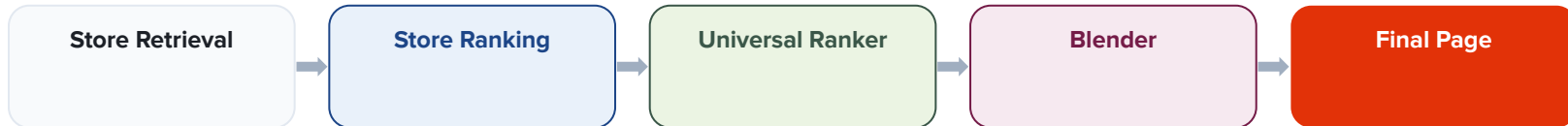
Stage 1: Multi-Level Ranking Stages

- 1. Carousel Retrieval & Capping:** Determines initial eligible containers.
- 2. Carousel Container Ranking:** Prioritizes store vs. item collections.
- 3. Item Ranking:** Determines specific items nested inside those carousels.

Stage 2: Downstream Post-Processing

- ✓ **Deduplication:** Enforces strict deduplication across separate containers.
- ✓ **Visual & Catalog Diversity:** Assures balanced merchant and category diversity.
- ✓ **Sponsored Blending:** Dynamically slots paid content to preserve coherence.

DoorDash: Staged Ranking Across the Whole Page



Universal Ranker: Vertical, Multi-Task

Compares heterogeneous page units against one another.

- ✓ **Vertical ranking:** store rows, carousels and item collections on a shared scale.
- ✓ **Multi-task:** clicks and same-day conversion, plus delayed labels such as ordering from a vertical within 7 days.
- ✓ **Session-aware:** a transformer over recent browsing lets the page react to the behaviour in this session..

Blending Relevance with Exploration

final score = [ranker score × business weight + exploration bonus]

A learned ranking core wrapped in business weights and an exploration bonus.

Two Ways to Compute the Bonus

- ✓ **UCB** — a bonus sized to how little we've seen this option; shrinks as impressions accumulate.
- ✓ **Thompson Sampling** — sample from a posterior belief about each option's reward, then rank by the sample.

Hard Constraints After Scoring

- ✓ **Ad placement** — required spacing between sponsored slots plus quality thresholds.
- ✓ **Deduplication** — the same item or merchant can't reappear across page surfaces.

Models estimate relevance → the blender applies weights and exploration → constraint systems finalize the page

Exploration Needs Constraints

The Guardrails

- ✓ **Capped bonuses** so exploration can never dominate the predicted value.
- ✓ **Partial traffic** — only a fraction of requests get the boosted experience.
- ✓ **Minimum impressions** before an option is eligible for a bandit bonus.
- ✓ **Warm start** puts new carousels in middle positions, not straight to the top.

Three Exploration Methods

Fixed Slot

Random cold-start position New content starts in a safe, non-top position

Traffic Slice

1% fully random ordering A small slice of sessions sees a fully randomized page

Bandit Bonus

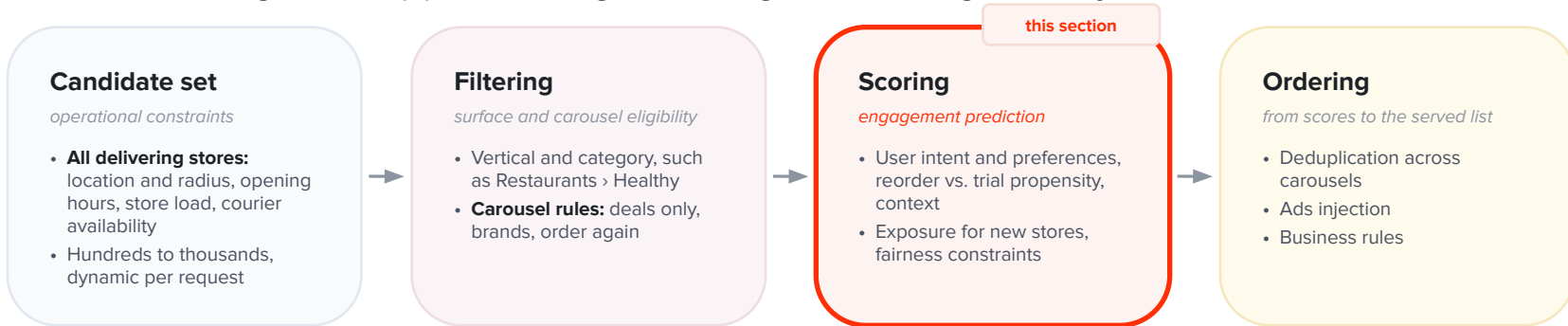
UCB Blend an uncertainty bonus into the score so new content earns, boost.

Advanced Ranking Architectures

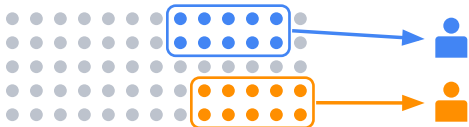
Presenters: Marcel Kurovski

Local constraints build the candidate set; scoring orders it

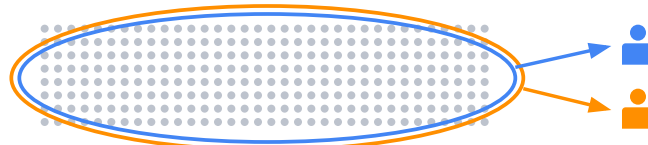
Where store ranking sits in the pipeline. Filtering and ordering frame it; scoring is the subject of this section.



Corpus size and how it clusters for different use cases (schematic)



Delivery: hundreds of thousands of stores; each user reaches a local cluster



YouTube: billions of videos; nearly all are candidates for every user

Scoring is the heart of store ranking. Physics and operations produce the candidate set; only very dense cities may call for a retrieval step. Ordering still moves stores after scoring: ads, deduplication, and business rules act on the scored list.

From DLRM via hybrids to transformer-centric ranking

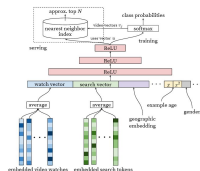
Three families, one direction: modeling the user as a behavioral sequence becomes increasingly important.

ESTABLISHED

DLRM

- Two-tower or MLP over dense and sparse features; users and items as ID and content embeddings
- Little to no sequential user modeling

YouTube DNN (2016) · Wide & Deep (2016) · DLRM (2019) · Uber Eats two-tower embeddings (2023)



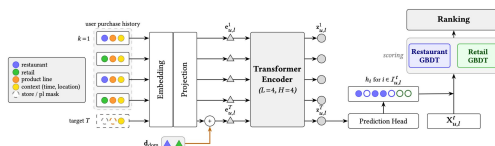
classic, well established

PRODUCTION TODAY

Hybrid

- A transformer encodes the user's action sequence; an MLP or GBDT combines it with tabular features
- Sequential modeling without giving up the feature stack
- **Wolt:** the encoder logit is one GBDT feature

BST (Alibaba 2019) · TransAct (Pinterest 2023) · UVR (Wolt 2026)



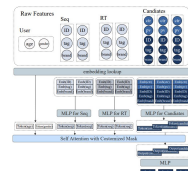
where Wolt ranks today

RESEARCH TODAY, NEXT STEP

Transformer-centric

- **Everything is a token:** user, past actions, real-time events, candidates
- One target-aware attention stack scores all candidates of a request in a single pass
- Quality scales with sequence length and depth; compute alone does not predict it

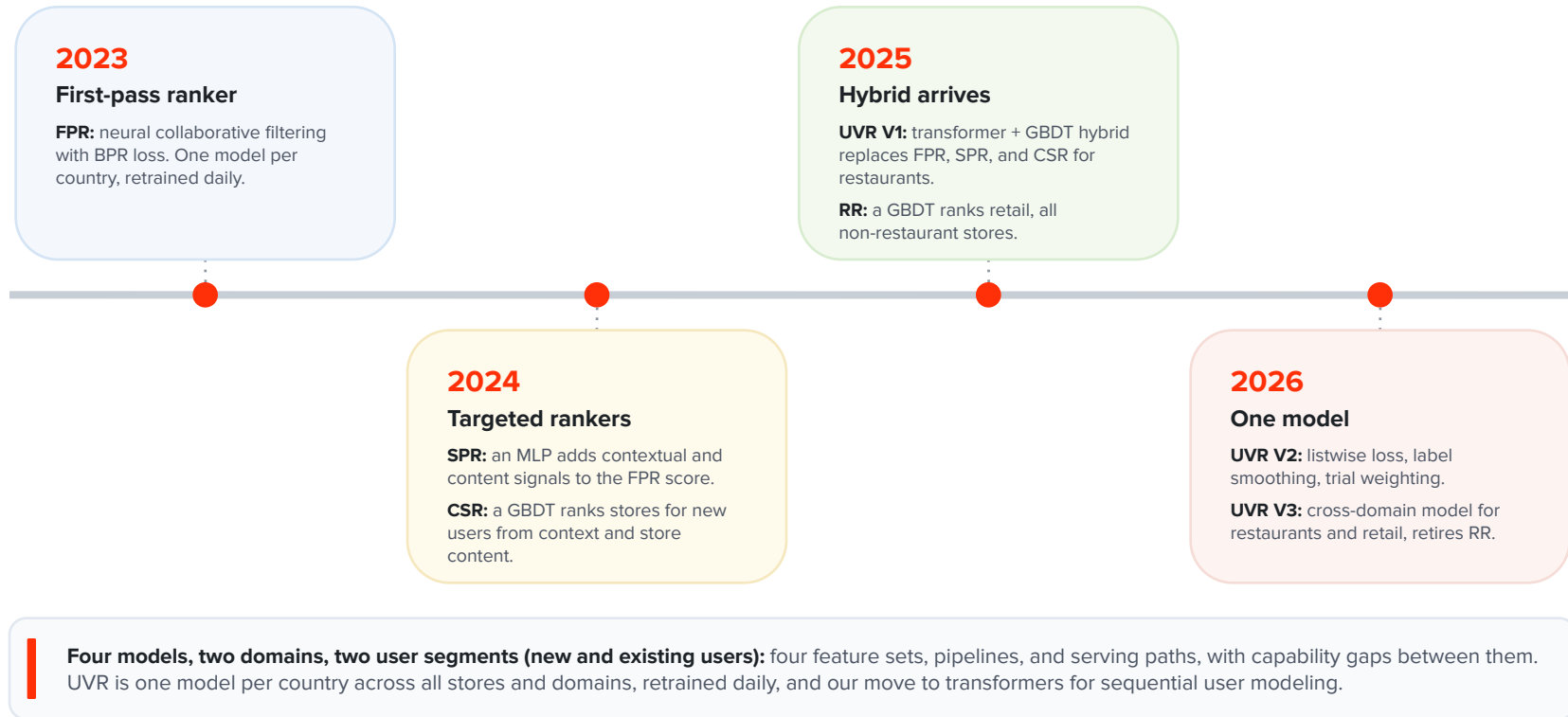
HSTU (Meta 2024) · MTGR (Meituan 2025) · LLaTTE (Meta 2026) · Uber Eats (2026)



where research and the next production step go

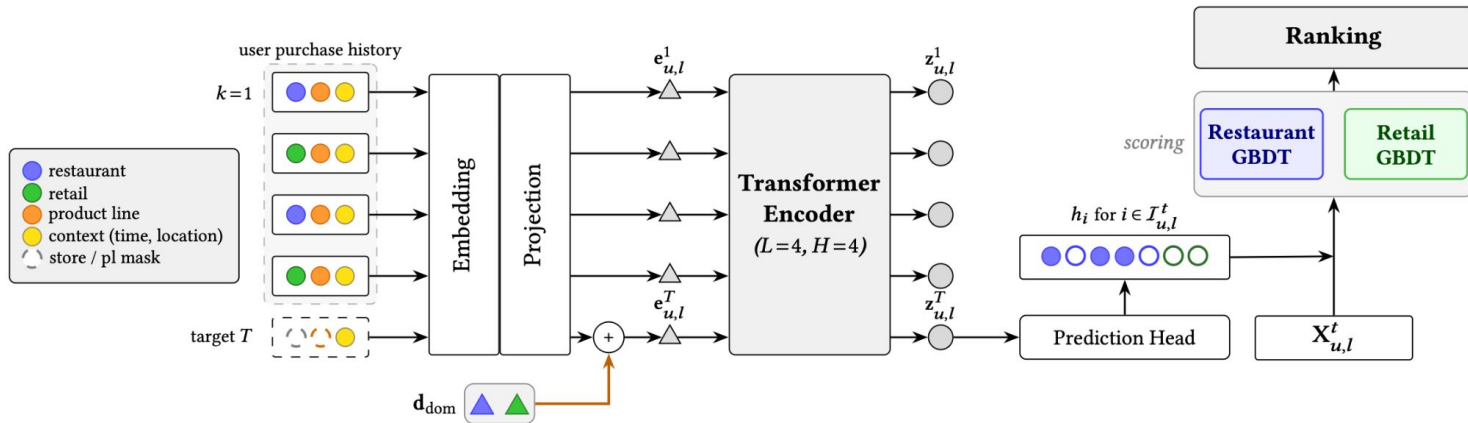
Four Wolt rankers became one universal ranker

Store ranking split by domain and user segment until 2025. UVR consolidated the landscape in three releases.



UVR cascades a sequence transformer into a GBDT ranker

Universal Venue Ranker. One forward pass scores the whole slate; the GBDT adds content, context, and user–store features.



Sequence classifier

- Bidirectional encoder over the purchase sequence (BERT4Rec backbone); only the target step is masked
- **Target step keeps what is known at request time:** time, H3 location cell, requested domain
- One pass yields a logit per store; candidate logits are gathered

GBDT ranker on top

- **CatBoost:** the logit as one scalar feature plus store content, context, and user–store features
- Listwise loss (YetiRank, NDCG) with clicks as graded relevance
- One ranker per domain, since feature sets differ

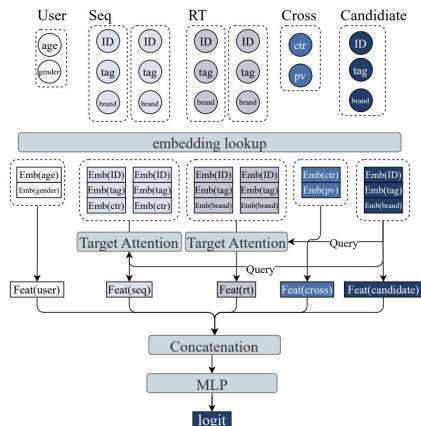
Trained in sequence, served global

- **Two stages on disjoint slices:** classifier prefixes end two days before the ranker's validation and test days
- One model per country across all stores and domains, retrained daily
- Local delivery constraints at inference; p99 about 60 ms on CPUs*

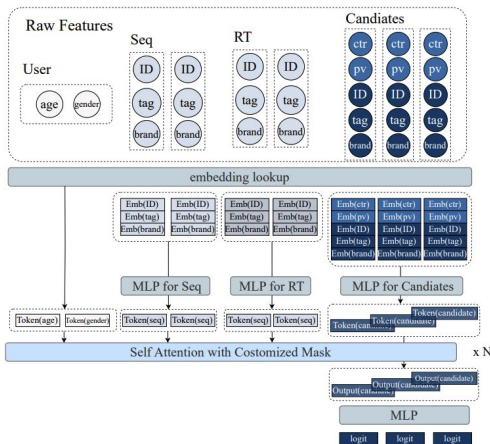
Transformer-centric models turn every input into a token

Same raw features, wired differently: attention as a feature extractor for an MLP, or one attention stack over all tokens.

Hybrid: DLRM with target attention



Transformer-centric: MTGR



Hybrid

- One forward pass per candidate; cost grows with the slate
- Sequence attention, then concatenation with tabular and cross features into an MLP or GBDT
- **Scaling is capped:** the MLP recomputes the sequence for each candidate

Transformer-centric

- User, actions, real-time events, and candidates are tokens of one sequence; one pass per request
- Cross features stay inside the candidate token; per-token-type normalization aligns the groups
- A mask lets candidates see the user, never each other; compute scales with depth and length

Figures: Han et al. (2025), MTGR: Industrial-Scale Generative Recommendation Framework in Meituan

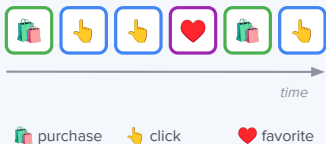
Where UVR stands: a hybrid with a bidirectional encoder that scores all candidates in one pass, but it is not target-aware, sees purchases only, and runs on daily batches. Each of these gaps is one of the advances on the next slides.

Grow the input sequence: richer, deeper, longer, steerable

Four ways the transformer-centric shift changes what a user sequence contains. UVR today: purchases only.

Richer

more action types

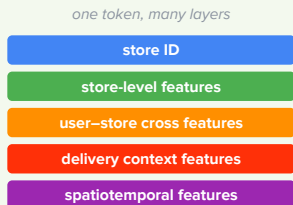


Clicks and favorites join purchases. An action-type embedding per token lets the model tell them apart and read intent from the mix.

PinnerFormer, TransAct (Pinterest) · Uber Eats: clicks and orders

Deeper

more features per token

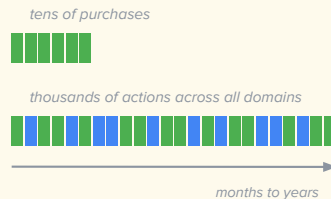


Each token carries store-level, user-store cross, delivery context, and spatiotemporal features, so a store token means more than its ID.

MTGR: cross features in the candidate token · LLaTTE: content features enable scaling

Longer

more history



From tens of purchases to thousands of actions across restaurants, grocery, and retail. Length shows the steepest scaling slope.

HSTU: up to 8k tokens · LLaTTE: gains up to 5k events

Steerable

stores and words together



Semantic IDs ground stores in content; a fine-tuned LLM reads store and language tokens interchangeably, so preferences can be stated in words.

TIGER, Singh et al. (2023) · Spotify (2025) · Generative Recommendations

Each dimension adds signal a purchase-only sequence misses. Richer and longer sequences help sparse users most; deeper tokens carry the feature stack into the transformer; steerable sequences make the ranker controllable in language.

Target-aware transformers score the slate in one pass

What HSTU, MTGR, and the Uber Eats ranker agree on, and what the delivery domain adds.

Target-aware

Candidates enter the sequence as tokens and attend to the whole history: HSTU interleaves items and actions, MTGR and Uber Eats append candidate tokens. UVR encodes the user once, independent of the candidate.

Multi-candidate, one pass

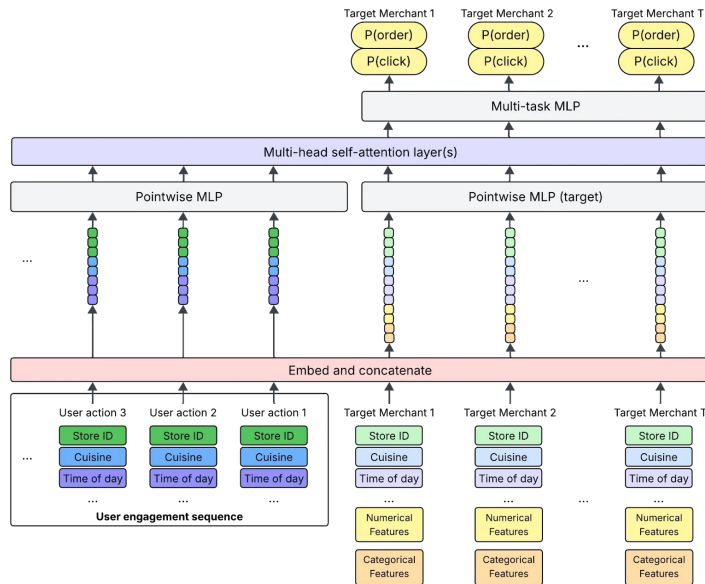
All candidates of a request form one sample. HSTU micro-batches them against a cached history; MTGR's mask keeps candidates from seeing each other. Per-candidate cost falls to roughly $1/T$ for T candidates.

Multi-task heads

One trunk, several heads: click and order (Uber Eats), CTR and CTCVR (MTGR). For delivery: trial and reorder as separate heads, and a follow-up-purchase head for post-order surfaces.

Real-time and scalable

Near-real-time action streams feed the sequence (Uber Eats: lag from days to seconds). Quality follows a power law in compute and length (HSTU, LLaTTE); LLaTTE moves the heavy user model off the request path.



Restaurant ranking at Uber Eats. Action tokens (store, cuisine, time of day) and target-store tokens share one attention stack; a multi-task MLP emits order and click probabilities per target. Diagram after Chen et al. (2026).

Delivery twist: candidates are local and few, so a single pass over the whole slate is cheap; physical constraints stay outside the model.

Ordering ends with deduplication, ads, and business rules

■ Homepage feed after scoring

Raw scoring output per carousel, before ordering. Scores are illustrative.

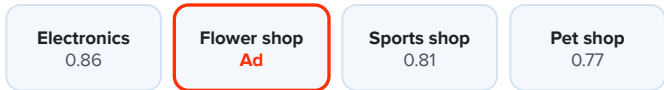
Recommended for you · restaurants and new verticals



Lunch near you · restaurants



Shopping deals · retail



restaurant grocery retail ad slot duplicate

Ordering next: keep one Burger joint, check spacing and quality of the ad slots, apply business rules, then rank the carousels.

■ Store Deduplication

Ensures repeat store entries are eliminated across different carousels and page sections. This rule restricts duplicate stores globally to diversify the results and to maximize scroll depth.

▲ Ads Injection

Dynamically slots paid store content natively within the home screen feed layout. This process enforces strict ad-quality thresholds, mandatory spacing gaps, and coherence filters.

◆ Business Rules

Applies sponsored placements, seasonal marketing boosts, and liquidity buffers. It also implements temporary exposure boosts for newly onboarded local stores to stabilize cold-start discovery.

Four things to carry into generative recommendations

- 1** **Candidates are local and few; store ranking is scoring plus ordering.** Physical constraints replace retrieval, so every candidate can be scored in one pass and heavier models per candidate are affordable. ↔ **whole-page optimization**
- 2** **Consolidation is an architecture decision.** One hybrid replaced four rankers: the transformer brings the sequence, the GBDT keeps the tabular feature stack, one model per country covers all stores and domains.
- 3** **Trial versus reorder is a tunable trade-off.** Trial weighting in UVR steers toward first orders at new stores, and it lifted them without a conversion cost.
- 4** **Next: target-aware, multi-task, real-time transformers** over richer, deeper, longer, and steerable sequences. → **Semantic IDs, generative retrieval**

Up next: generative recommendations. Raghav continues where the steerable sequence ends: Semantic IDs as catalog-native item addresses, decoding collections and candidates under marketplace constraints, and where generation enters a stack that still has to respect delivery radius and opening hours.

GENERATIVE RECOMMENDATIONS

From scoring a fixed candidate set to decoding context-aware collections and catalog-native item addresses.

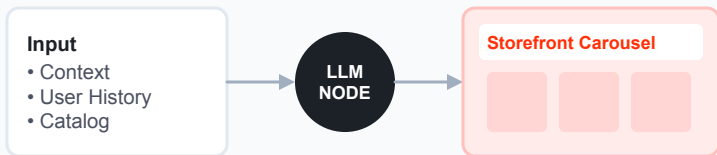
Presenter: Raghav Saboo

Generation enters the stack in two places

Generate the experience shown to the consumer—or generate the catalog candidate itself.

01 · EXPERIENCE GENERATION

Decode a collection specification



The model constructs structured representations directly, outputting metadata and design rules dynamically.

- The model creates intent, title, retrieval constraints, and diversity controls

OR

02 · GENERATIVE RETRIEVAL

Decode a semantic item ids

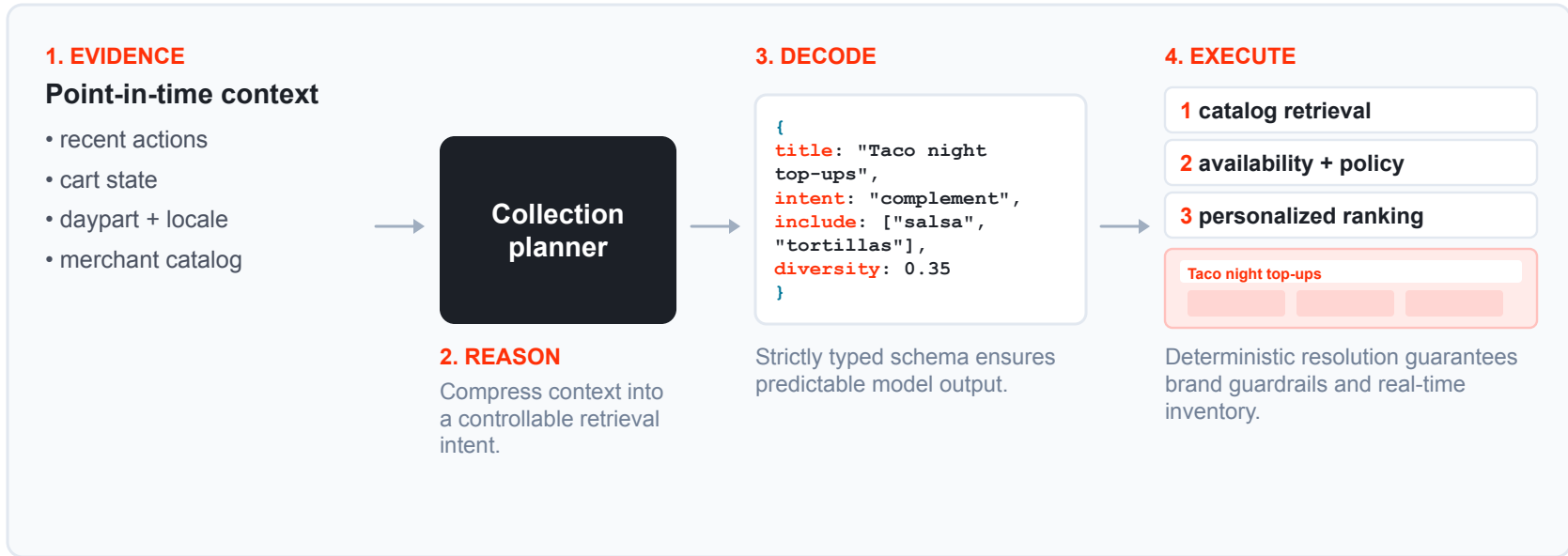


Addresses are structured hierarchically, allowing direct indexing of the entire item catalog.

- The model autoregressively emits a valid Semantic ID, replacing nearest-neighbor lookup with token decoding.

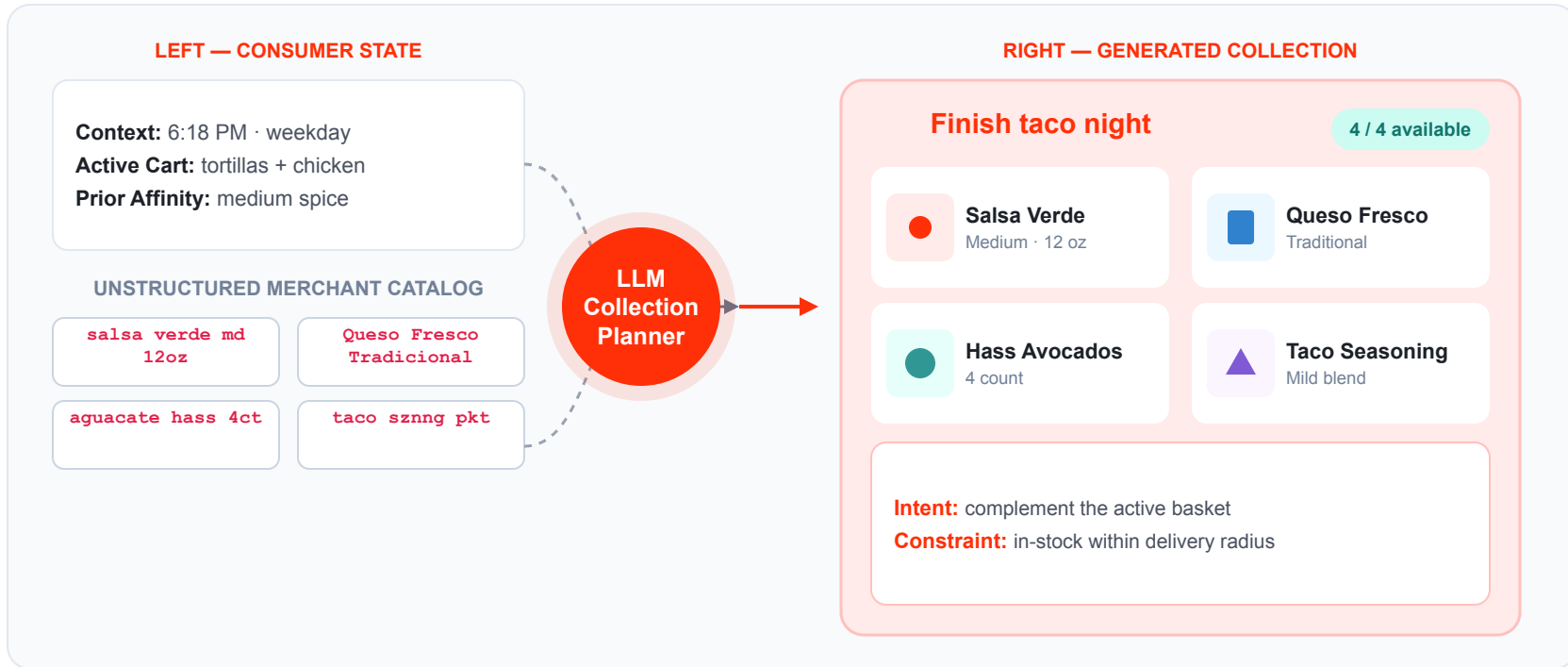
Generate a plan; resolve products deterministically

A production LLM produces a typed collection spec that retrieval and ranking can safely execute.



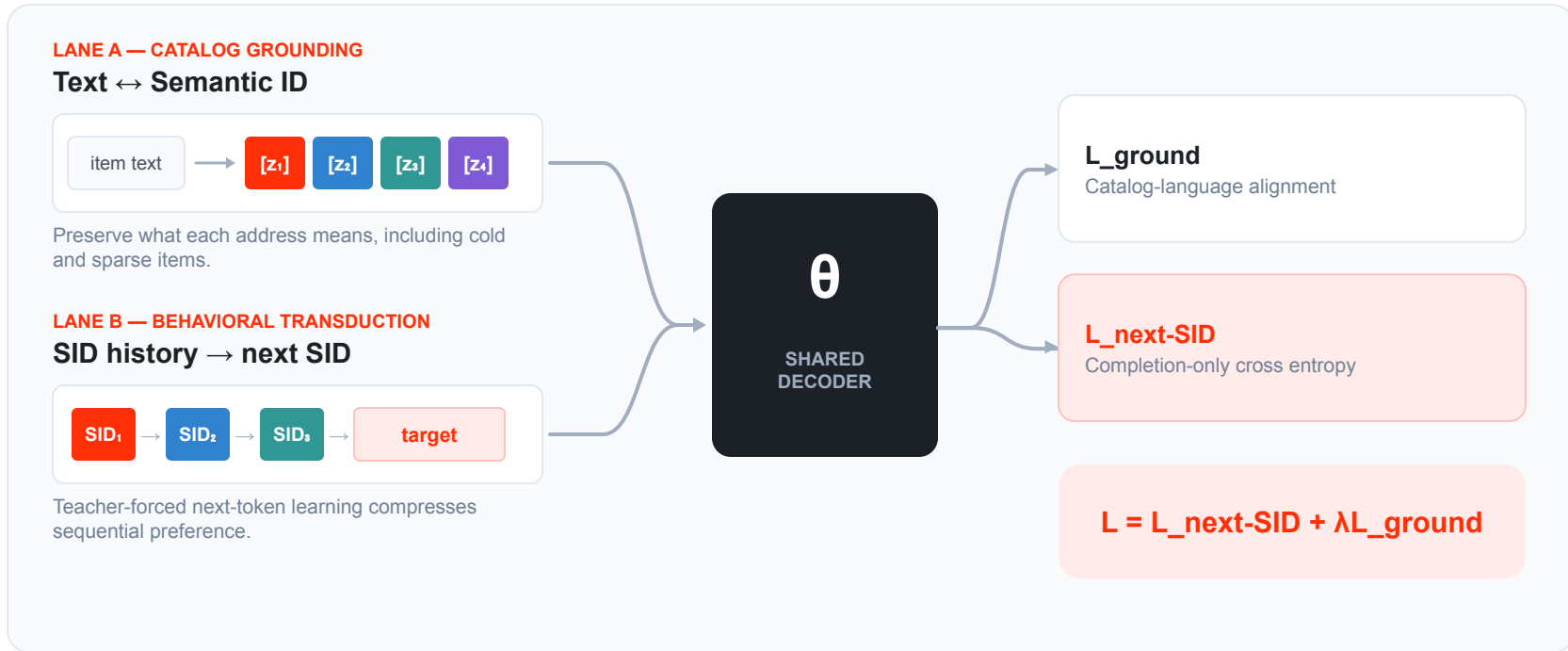
Noisy catalogs become coherent, local collections

The LLM bridges language, taxonomy, and behavioral context; deterministic retrieval keeps every item real.



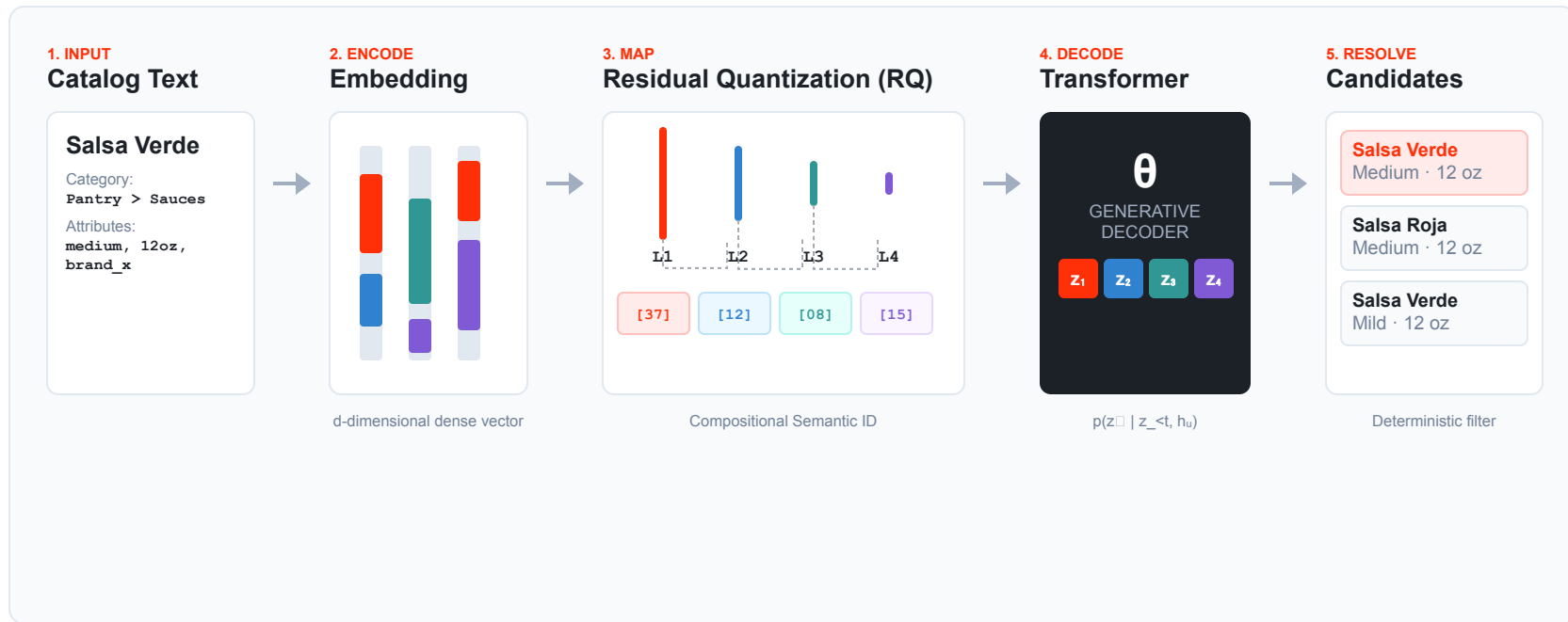
Ground the vocabulary, then learn behavior in token space

Two supervision lanes align catalog semantics with next-item preference inside one decoder.



Generative retrieval turns catalog items into tokens

Semantic IDs make item addresses compositional, so a sequence model can retrieve by autoregressive decoding.



Compile marketplace constraints into the decoder

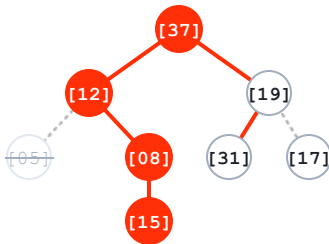
A request-specific trie converts locality, inventory, and policy into legal token continuations.

1. ELIGIBLE CATALOG Marketplace Filters

- ✓ delivery radius
- ✓ store open now
- ✓ item in stock
- ✓ policy + business rules

Dynamic request execution context

2. PREFIX TRIE Constrained Search Tree



$V'_t = \text{children}(\text{prefix}) \cap \text{trie}(\text{eligible})$

3. CONSTRAINED BEAM Deterministic Decoding

37-12-08-15 -0.42

37-19-31-24 -0.71

37-19-17-06 -0.96

Surviving legal continuations

Trie-constrained decoding guarantees catalog-valid addresses.

More expressive generation needs explicit catalog grounding

The generated output determines how we connect it to real, eligible products.



Broader output needs interpretation

Collection plans generate structure first. They require a downstream catalog retrieval step to resolve candidates into real, available products.

Semantic IDs preserve decoding control

Direct token generation uses structured item codes. Prefix trie-constrained decoding guarantees that every generated token path is valid in the catalog.

Grounding preserves marketplace control

Expressive generation works best when grounded explicitly. Trie-constrained decoding guarantees catalog-valid addresses without losing marketplace control.

Session 5:

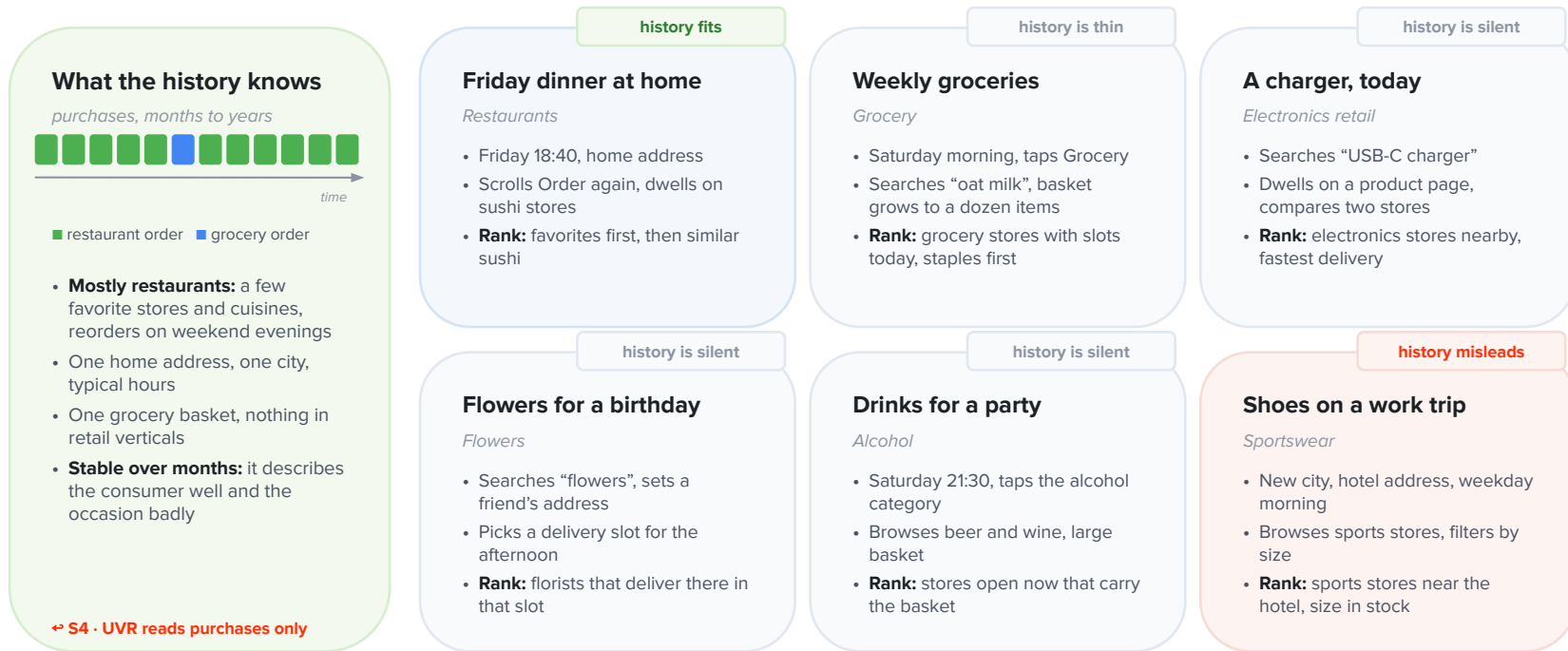
Future Directions and Open Problems

Real-time personalization · Agentic recommenders · Open discussion with the audience

Presenters: Raghav Saboo, Paavo Camps, Marcel Kurovski

Long-term preferences miss what this session is about

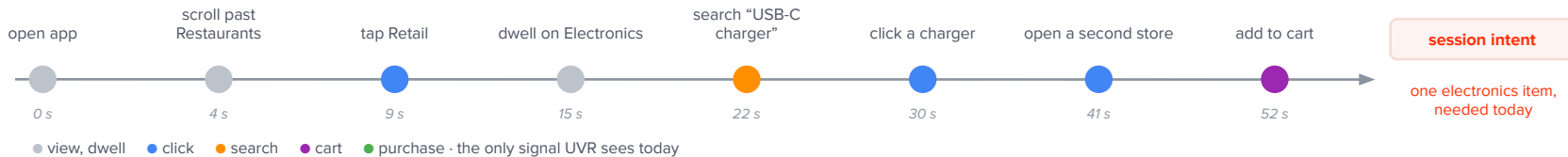
One consumer, six occasions. Months of restaurant orders say a lot about the first one and little about the rest.



The history describes the consumer. The session describes the occasion. Ranking inside the session needs both, and the second one arrives while the user is still scrolling.

Read session intent from signals that arrive in seconds

Dwells, scrolls, clicks, searches, and cart events form a second sequence next to the purchase history.



Sequential models over sessions

session-based recsys, real-time sequences

- **Two sequences per user:** purchases over months in daily batches, actions over seconds from a stream
- GRU4Rec and SASRec started session modeling; TransAct pairs a real-time action sequence with a long-term user embedding
- **Uber Eats:** feature lag from days to seconds

↔ S4 · real-time and scalable

Transformers that read both at once

history and session as one token sequence

- An action-type embedding tells a dwell from a click from a purchase; intent emerges from the mix
- Target-aware attention reads the same history differently for a sushi store and a charger store
- HSTU and MTGR carry real-time tokens next to the long-term ones

↔ S4 · richer, target-aware

Semantic IDs grounded in language

words and items share one vocabulary

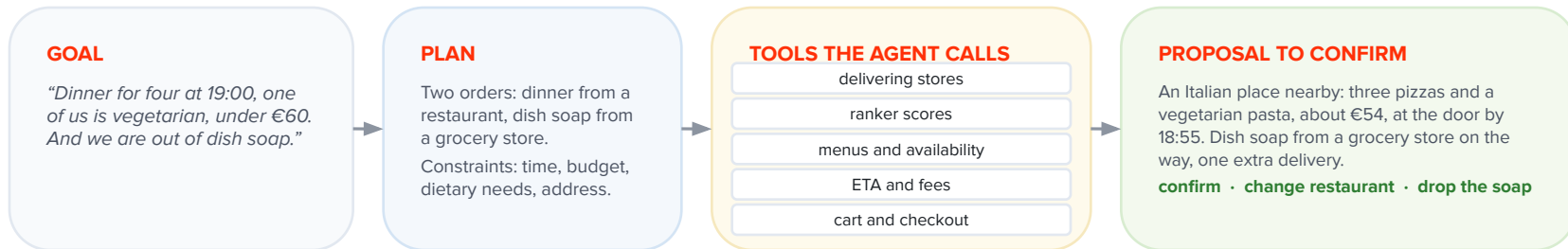
- A search "USB-C charger" and a tap on Electronics share tokens with the products they point to
- Generalizes to a vertical where the user has no history at all
- The steerable sequence from Session 4 becomes the input of in-session ranking

→ generative recommendations

Open problems. A streaming feature path and a heavier model inside the same latency budget · the first events of a session carry the least signal · session-level offline metrics and A/B designs · vertical switches are rare in the training data · long-term preferences still have to win when the session says nothing.

An agent turns a stated goal into a confirmed order

The consumer delegates a goal. The agent plans, calls the platform's systems as tools, and returns a proposal to confirm.



In-app assistant

the platform owns the whole flow

- Same candidate set and ranker as today, with a planner on top
- Explanations, confirmation, and trust become part of the UI
- Every step stays observable and testable

Our implementation: Ask DoorDash

Third-party assistants

catalog, availability, and ranking exposed as tools

- General assistants call the platform, for example over MCP*
- **The platform ranks for an agent reader:** one shot, structured, no carousel, no position bias
- Who owns the ranking, the platform or the agent?

The recommender becomes a tool

a contract, in both directions

- **In:** the goal and its constraints such as budget, dietary needs, time, address
- **Out:** calibrated scores, availability, fees, ETA, and reasons a planner can verify

↔ S4 · advanced ranking architectures

The interface moves from a ranked list to a delegated goal. Most of the stack stays. Its outputs have to become something a planner can consume, verify, and explain.

* MCP: Model Context Protocol, an open protocol that connects assistants to tools and data.

Is the agentic recommender a new system or a new UX?

Two open questions for the discussion, with our current reading. Conversational recommenders are the closest relative.

	Conversational RecSys	Agentic RecSys
Who drives	the user, turn by turn	the agent, until the goal is met
User gives	preferences and critiques	a goal with constraints
System returns	a list, refined in dialogue	a plan and an executed action
Scope	one recommendation task	several tasks, several stores
Recommender is	the whole system	one tool among many
Runs in	the platform's own UI	in-app or an external assistant
Evaluated on	list quality per turn	goal completion and trust
Lineage	CRS research, since about 2018	LLM agents, since about 2023

New system or new user experience?

our reading, open for discussion

- **The front changes most:** a goal instead of a query, multi-step proposals, confirmation, trust
- **Behind it, real system work:** tool contracts, ranking for an agent reader, constraint handling, evaluation by task completion
- The core ranker stays. Its outputs change.

What sets it apart from conversational RecSys?

our reading, open for discussion

- **Delegation and action:** a CRS refines a list with the user; an agent plans, calls tools, and acts, at times without a turn from the user
- Dialogue becomes optional; a goal statement is enough
- Both share preference elicitation and explanation

For the discussion: what would you let an agent order without asking? Who owns the ranking when the agent is someone else's? And which of today's offline metrics survive a reader that never scrolls?

Open discussion: pick any topic of the day

Every topic on the agenda is fair game. Three prompts to start, and the hallway afterwards.

Part I: Foundations & Challenges

08:30 - **Delivery Platforms as Multi-Sided Marketplaces**
09:00

- Trade-offs between consumers, merchants, and couriers
- Mapping offline ML metrics to online business KPIs

09:00 - **Challenges of Recommendations in Delivery**
09:40

- Locality constraints, real-time stock, & cold-starts
- Multi-objective optimization & high-dimensional scale

09:40 - **Surfaces and Entities for Personalized Recommendation and Ranking**
10:00

- Discovery to checkout: homepage, stores, and cart
- Diversity of entities: individual dishes to carousels

10:00 - **Coffee Break (30 min)**
10:30

Part II: Solutions & Future

10:30 - **Production Solutions for Delivery Recommendations**
11:45

- Item Representation with Semantic IDs
- Retrieval with Two-Tower Embeddings
- Multi-Task Learning for Ranking
- Whole Page Optimization and Explore/Exploit Frameworks
- Advanced Ranking Architectures
- Generative Recommendations

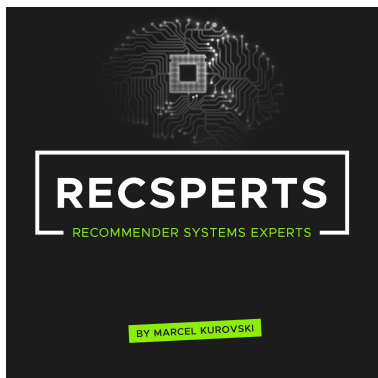
11:45 - **Future Directions & Open Problems**
12:00

- Real-time personalization
- Agentic recommenders
- Interactive Q&A and audience discussion

Prompts: Where does delivery ranking need retrieval? · Which real-time signals paid off for you? · What would you let an agent order unasked?

Hear more on RECSPERTS

Two long-form conversations on delivery recommendations. Scan a code to listen. Leave a subscription and review ;)



recsperts.com



Episode #32 · RecSys in the Delivery Industry at Wolt

with Sasha Fedintsev, Wolt · May 2026

- Locality constraints and repurchase patterns
- From collaborative filtering to transformer-based ranking
- Evaluation, bias, and why item ranking is harder than store ranking



scan for episode 32



Episode #34 · From Consumer Memory to Semantic IDs: Generative RecSys for Quick Commerce

with Raghav Saboo, DoorDash · September 2026

- Affordability, familiarity, and novelty in quick commerce
- LLM memory blocks as consumer and item profiles
- Semantic IDs for item ranking and query reformulation



scan for episode 34

We are hiring, at Wolt and at DoorDash

All open roles are on the career sites of both brands. Scan a code or follow the link.



Careers at Wolt

Open roles across our teams and locations, including machine learning, data, and engineering.



careers.wolt.com



Careers at DoorDash

Open roles across our teams and locations, including machine learning, data, and engineering.



careersatdoordash.com

Talk to us today or reach out afterwards. Both teams work on the problems from this tutorial: personalized ranking, experimentation, and marketplace decisions across restaurants, grocery, and retail.

References (1 of 2): Advanced Ranking Architectures

Session 4. Papers, posts, and tools cited on the slides of this section.

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References (2 of 2): Future Directions

Session 5. Papers cited on the slides of this section. Other sections append here.

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Thank you to our contributors



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Thank you

Questions and comments are welcome, now and afterwards.

Recommender Systems in Delivery Platforms · RecSys 2026 Tutorial

Presenters: Marcel Kurovski · Paavo Camps · Raghav Saboo

careers.wolt.com · careers.doordash.com

Backup

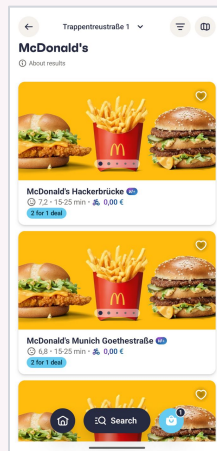
Preference forms at the brand, feasibility lives at the store

Consumers think in brands where they exist. Fulfilment comes from one store, branded or independent.

01

What the consumer perceives: the brand, if there is one

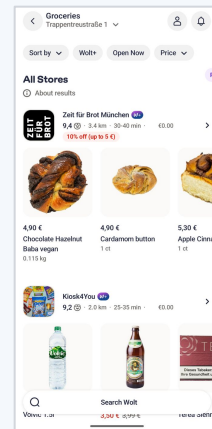
- **A chain, a franchise, a dark-store brand:** “I like this place”. The preference travels with the consumer across cities.
- **McDonald's in Munich (right):** one brand, several stores with their own rating and ETA. Wolt Market: one brand, many dark stores.
- **Many stores carry no brand:** independent mom-and-pop restaurants and shops. They surface on their own merits, so brand is one signal among several.



02

What the marketplace fulfils from: the store

- One physical store with its own opening hours, delivery radius, courier supply, stock, prices, and ETA to this address.
- Availability, ETA, and price are store-level facts and change by the hour.
- The same brand can be excellent in one city and undeliverable at this address.
- **Relocation:** brand affinity survives, store affinity is lost. ↔ S2 glocal, cold start



The model joins brand-level preference, where a brand exists, to store-level feasibility. Get the join wrong and you either drop a known preference or promise something undeliverable.

A store card is itself a mini-recommendation: the dishes or products chosen as its shopfront (right: Zeit für Brot with three bakery items) are an item ranking nested inside the store ranking.

Purchase-type weights set the trial–reorder operating point

Trial: a user's first order from a store. Reorder: a store they know. UVR makes the balance between them explicit and tunable.

The tension

Users return to trusted stores out of habit, yet expect to find stores they have not tried. Every purchase is typed as **cold-start, new** (trial), or **recurring** (reorder), and the model is steered by type.

The levers

- **Classifier:** per-sample weights new > cold-start > recurring, label smoothing; checkpoint picked on trial MRR with reorder and cold-start as guardrails
- **Ranker:** per-session weight ratio new/recurring (1.5, 2.0, 3.0 in the plot) and the number of sampled negatives move along the frontier
- **Constrained problem:** maximize trial quality subject to a bounded reorder regression

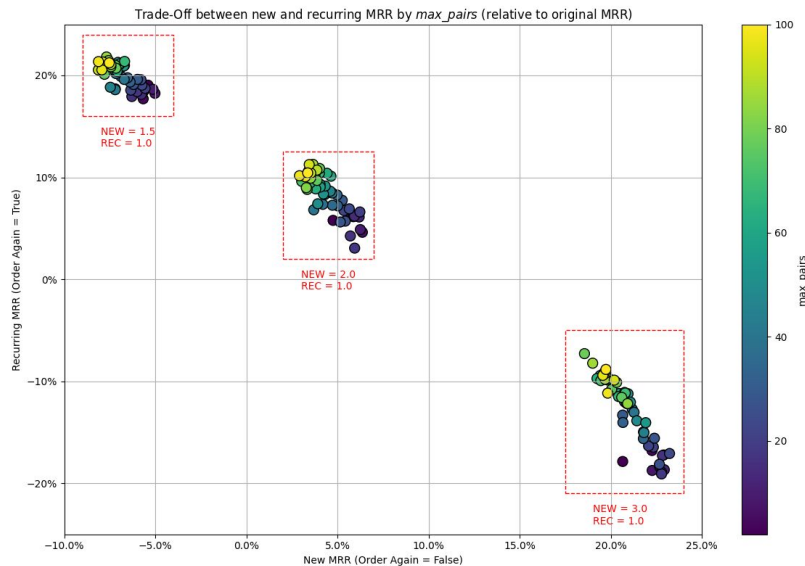
Online outcome, three consecutive A/B tests

V1 +5.5% Merchant Trial Rate · +0.16% Global CVR

V2 +0.45% Merchant Trial Rate

V3 +1.31% Merchant Trial Rate · retail domain

More trial without a conversion cost. First orders at new stores relate to higher order frequency and better retention.



Ranker ablation with the classifier fixed. Relative MRR uplift over production on trial purchases (x) against reorders (y). Three clusters are the three weight ratios; the color is the number of sampled negatives. Moving right buys trial and costs reorder.

Where we operate: the setting with the largest trial uplift whose reorder regression stays bounded. The reorder cost did not surface in the blended conversion metric.

Outputs become calibrated, sliced by intent, and reusable

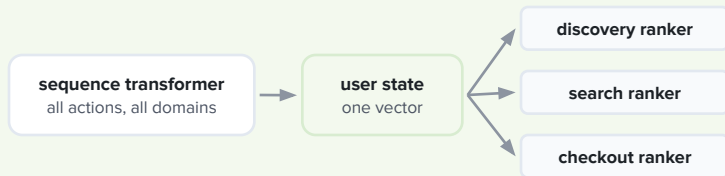
Where model outputs are heading: probabilities other systems can consume, and a user state other rankers can share.

Calibrated probabilities

- From uncalibrated, uninterpretable ranking scores to calibrated pCTR and pCVR per candidate
- **Sliced by intent:** separate estimates for trial and reorder, so the operating point becomes a per-session decision instead of a training-time weight
- **Consumable downstream:** ads auctions, blending of organic and sponsored slots, and whole-page optimization need probabilities on one scale

Distilled user state

- The sequence transformer emits a compact user embedding once, upstream of any request
- **Shared as a feature by discovery, search, and checkout rankers:** the sequence signal without paying its cost per surface
- **Precedent:** PinnerFormer embeddings inside Pinnability; LLaTTE's asynchronous upstream user model



Why the output matters as much as the input. A ranker whose scores mean nothing outside its own list cannot be blended, audited, or reused. Calibrated, intent-sliced probabilities plug into ads, blending, and page construction; a distilled user state lets every ranker share the sequence signal.

SCRATCH · drop the 22 PNGs from presentation/screenshots_wolt_cropped onto this slide.
Any position, any size. Then run build_part2.py --place-shots. Delete this slide once the images are placed.

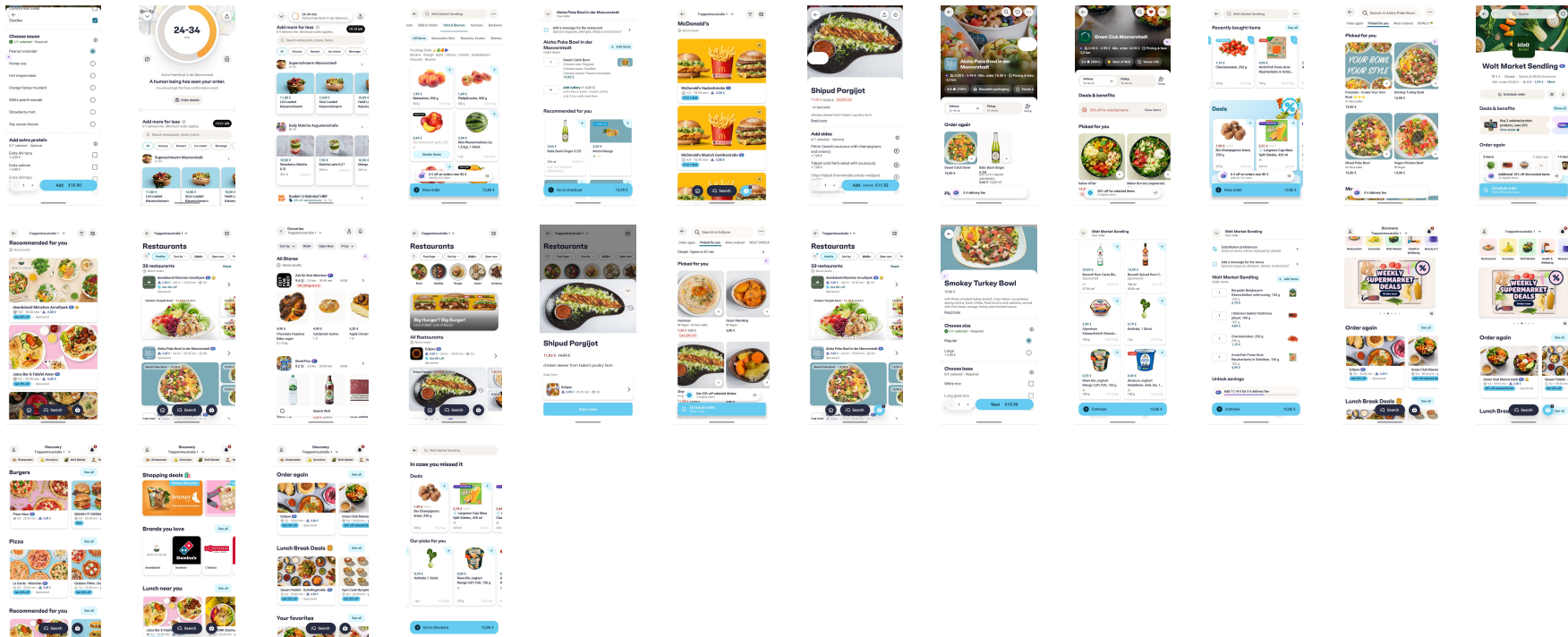


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